

Phenix Frequently Asked Questions

8 documentation pages

Generated from local Phenix documentation

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Cryo_fit2 FAQ

cryo_fit2_FAQ.html

Contents

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- How can I provide gaussian filtered maps?
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- My model is shifted after phenix.model_idealization.
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- "Sorry: Crystal symmetry mismatch between different files."
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- When cryo_fit1 and cryo_fit2 are useful?
- When cryo_fit2 is useful?
- When cryo_fit2 is more useful than cryo_fit1?
- What to do after cryo_fit?

After cryo_fit2, secondary structures of my input molecule tend to be broken.

- Doo Nam observed that when he provided a little less ideal helix (i.g. one H-bond length within helix is rather long), then cryo_fit2 tends to "loosen/unwind" that "kink" area.

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For example, phenix.cryo_fit2 <user.pdb> <user.map> resolution=<reso>
secondary_structure.protein.distance_cut_n_o=4.5

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secondary_structure.protein.distance_cut_n_o=4.5

- Alternatively, a user can specify a lower map_weight_multiply.

For example, phenix.cryo_fit2 <user.pdb> <user.map> resolution=<x> map_weight_multiply=0.2.

- For example, `phenix.cryo_fit2 <user.pdb> <user.map> resolution=<x> map_weight_multiply=0.2`.
- Alternatively, a user can use `phenix.model_idealization` after `cryo_fit2` to restore broken secondary structures (if any).

How can I provide gaussian filtered maps?

- Many people in flexible fitting field know that often providing gaussian filtered maps results in better fitting.

For example, flexibly fit with a 4 angstrom filtered map. Then with the resultant atomic model and a 2 angstrom filtered map, do flexible fitting. Then with the resultant atomic model and a non-filtered map, do final flexible fitting.

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- Then with the resultant atomic model and a 2 angstrom filtered map, do flexible fitting.
- Then with the resultant atomic model and a non-filtered map, do final flexible fitting.
- UCSF Chimera can generate these filtered maps.

Tools -> Volume Data -> Volume Viewer -> Tools -> Volume Filter -> (Set Width, e.g. 4,2) -> Filter File -> Save map as -> (name a filtered map) -> Save

- Tools -> Volume Data -> Volume Viewer -> Tools -> Volume Filter -> (Set Width, e.g. 4,2) -> Filter
- File -> Save map as -> (name a filtered map) -> Save

How to extract a relevant map region?

- Question: I only want to fit the monomer because only one monomer of the trimer map is the best. Do you have any idea of how to generate the monomer map that `cryo_fit` can use?
- Answer: Please use `phenix.map_box` to extract a local map near entered atomistic model.

Alternatively, a user can try `phenix.map_to_model` instead.

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How to improve initial cc?

- `phenix.dock_in_map`, UCSF Chimera's 'Fit in Map (Tools -> Volume Data -> Fit in Map)', UCSF Chimera's manual fitting (using a mouse) often improve initial cross-correlation (cc) between map and model.

Note that these options do rigid-body fitting only. Therefore, it is useful for global fitting before `cryo_fit2`. When overall orientation is already well fitted, these often are not that useful for `cryo_fit2`.

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- UCSF ChimeraX's `isolve` may improve initial cc as well.

How to maximize cc?

- Generally, `cryo_fit2` which is followed by `phenix.real_space_refine` tends to maximize cc.

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- However, if there is a room/chance to idealize secondary structure further even after cryo_fit2, then cryo_fit2 + phenix.model_idealization + phenix.real_space_refine combination resulted in the maximum cc.

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- When the fitting needs to pass through a local minimum energetically for a perfect fit (for example, when hydrophobic interaction between helices hinders loop conformation change like or when initial cc is too low (e.g < 0.2) like or when local sidechain fitting hinders global backbone fitting like), running phenix.dock_in_map or UCSF Chimera's 'fit in map' before cryo_fit2 is recommended.

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- When fitting protein molecule to a map requires a significant conformational change (for example, half of the molecule is already fitted and the other half is not fitted like),

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enforcing stronger map_weight (higher map_weight_multiply) tends to fit better (obviously). For example, phenix.cryo_fit2 <user.pdb> <user.map> resolution=<x> map_weight_multiply=15 At this high map_weight_multiply, the secondary structures may be broken (slightly or seriously) right after cryo_fit2 running. However, phenix.model_idealization which is followed by phenix.real_space_refine perfectly restored ideal secondary structures and fitted very well like (cc_box: 0.96). However, too high map_weight_multiply tends to misfit totally. For example, phenix.cryo_fit2 <user.pdb> <user.map> resolution=<x> map_weight_multiply=1000 misfitted like (yellow: after cryo_fit2, green: after cryo_fit2 which is followed by model_idealization). Lowering many sigma values was expected to prevent this situation. However, lowering many sigma values prolongs cryo_fit2 running time too much. Alternatively, remodel loop with phenix.fit_loops helps cc boost. (yellow: user input, cc_mask: 0.43, cc_box: 0.58; pink: after fit_loops, cc_mask: 0.47, cc_box: 0.58) (green: after fit_loops & cryo_fit2, cc_mask: 0.83, cc_box: 0.90) (green: after fit_loops, cryo_fit2, model_idealization & real_space_refine, cc_mask: 0.92, cc_box: 0.97) Alternatively, HE_top_out=True sometimes (not always) boosts cc well even with moderately high map weight (confirmed with 6 benchmark cases). For example, phenix.cryo_fit2 <user.pdb> <user.map> resolution=<x> map_weight_multiply=10 HE_top_out=True HE_top_out=True uses top_out potential (rather than harmonic potential) to helix (H) and sheet (E) for more flexible fitting.

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HE_top_out=True
- HE_top_out=True uses top_out potential (rather than harmonic potential) to helix (H) and sheet (E) for more flexible fitting.

I have a SAXS based cryo-EM map.

- Please use map_weight < 0.2 for SAXS (small angle x-ray scattering) data based cryo-EM map A user can transform SAXS data into cryo-EM map (mrc/sit) by Situs SAXS However, this is nothing but central points by SAXS, although UCSF Chimera will visualize this "cryo-EM" map (mrc/sit) as regular cryo-EM map
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I have a very small molecule to fit.

- When Doo Nam tried to fit a very small molecule (i.e. 11 residues long like), both phenix.real_space_refine and cryo_fit2 couldn't fit to cryo-EM map.

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- The reason of these failures is that effect of sidechains is too large (in this kind of small molecule), so that global fitting is hindered.

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- Doo Nam confirmed that phenix.dock_in_map fits globally much better for this case.

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Is there a command line option for cryo_fit2 processing on multiple cores?

- (Please be advised that using multiple cores for cryo_fit2 is only useful when explore=True)
- Like other PHENIX applications, specify nproc.

For example, phenix.cryo_fit2 nproc=40 model.pdb model.map resolution=3

- For example, phenix.cryo_fit2 nproc=40 model.pdb model.map resolution=3
- If nproc is not specified, cryo_fit2 will use 'available number of cores'/3

My input (before cryo_fit2) and output (after cryo_fit2) pdb files have different molecule sizes

Although pdb text files show similar SCALES, input (before cryo_fit2) and output (after cryo_fit2) pdb files may show different molecule sizes in Pymol when Doo Nam used map that was made from phenix.map_box

Troubleshooting is ongoing.

In the meantime, please use UCSF Chimera to visualize bio-molecules instead.

My model is shifted after phenix.model_idealization.

- To prevent this, it is recommended to add map as well.

For example, `phenix.model_idealization <user.pdb> <user.map that was used for cryo_fit2>` (resolution information is not required)

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"Number of atoms with unknown nonbonded energy type symbols"

- When a user sees a message like "Sorry: Fatal problems interpreting model file: Number of atoms with unknown nonbonded energy type symbols: 40 Please edit the model file to resolve the problems and/or supply a CIF file with matching restraint definitions, along with `apply_cif_modification` and `apply_cif_link` parameter definitions if necessary.", please check proper namings in a pdb file. For example, Doo Nam observed this message when he provided XXX in residue names instead of ALA, TYR, PHE etc. To prevent this situation, he needed to provide sequence information when he ran `phenix.fit_loops`, For example, `phenix.fit_loops <pdb file> <mrc file> <fasta file> start=<> end=<>`

When a user sees a message like

- "Sorry: Fatal problems interpreting model file: Number of atoms with unknown nonbonded energy type symbols: 40 Please edit the model file to resolve the problems and/or supply a CIF file with matching restraint definitions, along with `apply_cif_modification` and `apply_cif_link` parameter definitions if necessary.",
- please check proper namings in a pdb file.

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"RuntimeError: cctbx Error: Miller index not in structure factor map"

- When a user sees above message, consider to enter a correct resolution.

For example, Doo Nam uses either EMDB reported resolution (preferred) or phenix.mtriage derived resolution to properly run cryo_fit2.

- For example, Doo Nam uses either EMDB reported resolution (preferred) or phenix.mtriage derived resolution to properly run cryo_fit2.
- When a user is not sure about correct resolution, often using a lower resolution works.

For example, when Doo Nam saw an error with resolution=3, then using resolution=5 often solved the problem.

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"Sorry: Crystal symmetry mismatch between different files."

- Sometimes, Doo Nam observed,

"Sorry: Crystal symmetry mismatch between different files. (378, 378, 378, 90, 90, 90) P 1 (103.95, 91.35, 89.25, 90, 90, 90) P 1" The first line (378, ...) shows unit cell parameters from model. The second line (103.95, ...) shows unit cell parameters from map.

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- Solutions

Just erase CRYST1 header information from an input pdb file. Then, provide that input file into cryo_fit2.

Cryo_fit2 will automatically extract CRYST1 header from map (both from original map and phenix.map_box derived map), and prepend (write at the first line) to the pdb file. (Alternatively) Just leave correct CRYST1 header only in an input pdb file. For example, when a wrong CRYST1 header exists above a correct CRYST1 header, above error message appeared.

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What is the theory behind cryo_fit2?

- See the cryo_fit2 theory

When cryo_fit1 and cryo_fit2 are useful?

- When a user needs to model dynamic conformational change.

For example, cryo_fit can model transition between map conformation 1 and 2. see movie generation by cryo_fit1. see movie generation by cryo_fit2.

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- see movie generation by cryo_fit1.
- see movie generation by cryo_fit2.
- When the initial atomistic model is different from the map.

When cryo_fit2 is useful?

- To simply fit to cryo-EM map, cryo_fit2 seems to be useful when initial cc_mask and cc_box < 0.6. When initial cc_mask and cc_box >= 0.6, phenix.real_space_refine tends to do similar fitting with faster speed.

When cryo_fit2 is more useful than cryo_fit1?

- When fitting requires a larger movement.

For example,

- For example,
- To use non-canonical chemical entities using phenix.eLBOW.
- When gromacs is not installed.

Unlike cryo_fit1 that uses gromacs, cryo_fit2 runs within PHENIX suite. Therefore, it does not require gromacs installation and is faster to execute.

- Unlike cryo_fit1 that uses gromacs, cryo_fit2 runs within PHENIX suite. Therefore, it does not require gromacs installation and is faster to execute.
- cryo_fit2 better fits than cryo_fit1 in 7 cases out of 10 cases (cryo-EM maps have 3~24 angstrom resolutions. They tie for 2 case, cryo_fit1 better fits for 1 case where artificially made cryo-EM map is used).

What to do after cryo_fit?

- If there is a room/chance to idealize secondary structure further even after cryo_fit, then run phenix.model_idealization which is followed by phenix.real_space_refine.

- Otherwise, just run `phenix.real_space_refine` to improve overall geometry (such as rotamers).

Cryo_fit1 FAQ

cryo_fit_FAQ.html

Contents

- cc values have kept decreased (or have not increased that much) during step 8
- How can I provide gaussian filtered maps?
- How can I use double digit/character chain ID pdb file?
- How long does it take to run cryo_fit?
- How to draw a figure with cc values?
- How to enlarge map box size?
- How to extract a relevant map region?
- How to generate and record movie?
- How to improve initial cc?
- How to maximize cc?
- I can't run phenix.superpose_pdb with cryo_fitted pdb file
- I see "Fatal error: A charge group moved too far between two domain decomposition steps. This usually means that your system is not well equilibrated" at my 8_cryo_fit step.
- I see "Fatal error: Incomplete ring in HIS50" at my 8_cryo_fit step
- I see "Fatal error: Number of grid cells is zero. Probably the system and box collapsed." at my 8_cryo_fit step.
- I see "pdb file cleaning is not done" at my step 1 (Make gro and topology file by regular gromacs)
- I see "Software inconsistency error" at my 8_cryo_fit step
- I see "state.cpt not found, step_8 may be full of stepxb_nx.pdb" at my 8_cryo_fit step
- I see "step 0 correlation coefficient: nan" at my 8_cryo_fit step.
- I see "step 0 correlation coefficient: nan" and "Range checking error: Explanation: During neighbor searching, we assign each particle to a grid based on its coordinates. If your system contains collisions or parameter errors that give particles very high velocities you might end up with some coordinates being +-Infinity or NaN (not-a-number). Obviously, we cannot put these on a grid, so this is usually where we detect those errors. Make sure your system is properly energy-minimized and that the potential energy seems reasonable before trying again." at my 8_cryo_fit step.
- I see "The domain decomposition grid has shifted too much" at my 8_cryo_fit step
- I see "The initial cell size (xxx) is smaller than the cell size limit (xxx), change options -dd, -rdd or -rcon, see the log file for details" at my 8_cryo_fit step
- I see "Too many LINCS warnings" at my 8_cryo_fit step
- I see "User's provided atomic model had 0.0 cc" in my cryo_fit.overall_log.
- My model is shifted after phenix.model_idealization.
- When cryo_fit is useful?
- What to do after cryo_fit?

cc values have kept decreased (or have not increased that much) during step 8

- Initial few steps of cryo_fit in step 8 may decrease cc values temporarily as an initial perturbation.
- However, if it keeps decreasing continuously (or do not increase that much), it may indicate two possible cases.

Case 1 <A user input pdb file has reasonable structural geometry> For this case, try below possible solutions. If cryo_fit still can't find better (higher) cc, then the initial correlation between user input pdb file and cryo-EM map is already high enough. Just run phenix.real_space_refine only and deposit. Doo Nam recommends to argue/claim in a paper like "Our fit is so high, even cryo_fit did not find higher level of fit than our initial fit". This bold recommendation is based on his limited observations so far. Therefore, if you have a better cc model (from manual fit?) than cryo_fitted model, we would appreciate if you can email us (jprajapati@lanl.gov, dgirodat@lanl.gov, doonam.kim@gmail.com). Case 2 <A user input pdb file has unreasonable structural geometry> For this case, although the initial fit between user's atomic model and map looks good, it is a fictitious fitting without consideration of ideal atomistic model geometry. Run cryo_fit to find decent fit to the cryo-EM map that restores/maintains reasonable structural geometry based on molecular dynamics forcefield (amber03.ff).

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- Case 2 <A user input pdb file has unreasonable structural geometry> For this case, although the initial fit between user's atomic model and map looks good, it is a fictitious fitting without consideration of ideal atomistic model geometry. Run cryo_fit to find decent fit to the cryo-EM map that restores/maintains reasonable structural geometry based on molecular dynamics forcefield (amber03.ff).
- There are possible solutions

Provide a higher resolution map (like better than 4.5 Angstrom) tends to find higher cc since there is a more chance of finding fittable regions. Cryo_fit calculates the gradient of cc. If cryo_fit is provided a giant cryo-em map with a tiny atomic model, then there is a large empty space (not filled). Therefore, the constraint forces for the model are very small. Consequently, these small forces are not helpful. 2-1. Re-run cryo_fit with an atomic model that fits the majority of the map. If you fit multiple atomic models into a symmetric map or do sequential fitting into a non-symmetric map, watch Tom Goddard's lecture (2013) 2-2. Re-run cryo_fit with only a relevant map region. See 'How to extract a relevant map region?' in this FAQ. If the initial model is not properly aligned to a map, see 'How to improve initial cc?' in this FAQ. A user may enforce a stronger initial map weight (e.g. emweight_multiply_by) manually. However, since cryo_fit automatically increases emweight_multiply_by if it doesn't find higher cc anyway, manual increase in map weight may not necessarily find higher cc.

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How can I provide gaussian filtered maps?

- See 'How can I provide gaussian filtered maps?' in cryo_fit2_FAQ

How can I use double digit/character chain ID pdb file?

- For a large molecule (like ribosome), it is typical to put a prefix to chain ID so that the ID is double digits/characters. Since gromacs cannot deal with this, cryo_fit automatically removes the prefix.
- If a user sees an error with single chain ID for phenix.real_space_refine, Doo Nam recommends to use

python <User

Phenix>/modules/cryo_fit/files_for_steps/9_after_cryo_fit/solve_duplicate_atoms_issue/add_all_prefix.py

- python <User
Phenix>/modules/cryo_fit/files_for_steps/9_after_cryo_fit/solve_duplicate_atoms_issue/add_all_prefix.py

How long does it take to run cryo_fit?

How to draw a figure with cc values?

- python <phenix_path>/modules/cryo_fit/files_for_steps/9_after_cryo_fit/draw_cc/draw_cc.py cc_record

If a user is using ssh linked linux, run 'set DISPLAY' to avoid the message of "Unable to access the X Display, is \$DISPLAY set properly?".

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- PHENIX GUI for cryo_fit shows this figure automatically.

How to enlarge map box size?

- One can know map size by

cryo_fit which reports a map size between step 6 and step 7 with codes of from iotbx import ccp4_map
ccp4_map = ccp4_map.map_reader(user_input_map) target_map_data = ccp4_map.map_data() EMAN2's
e2iminfo For example, e2iminfo.py <user>.mrc (note) As of EMAN2.2, the file extension should be mrc, not
ccp4 VMD which can visualize current map size as in this FAQ.

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- VMD which can visualize current map size as in this FAQ.
- Then, relion_image_handler can enlarge map box size.

For example, relion_image_handler --i <user>.mrc --new_box 370 --o <user_box_size_370>.mrc

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How to extract a relevant map region?

- See 'How to extract a relevant map region?' in cryo_fit2_FAQ

How to generate and record movie?

- See Generate_record_movie_by_cryo_fit1

How to improve initial cc?

- See 'How to improve initial cc?' in cryo_fit2_FAQ

How to maximize cc?

- Generally, as seen in Cryo_fit: Democratization of flexible fitting for cryo-EM, 2019, cryo_fit1 which is followed by phenix.real_space_refine tends to maximize cc.
- However, if there is a room/chance to idealize secondary structure further even after cryo_fit1, then cryo_fit1 + phenix.model_idealization + phenix.real_space_refine combination resulted in the maximum cc.
- When the fitting needs to pass through a local minimum energetically for a perfect fit (half of the molecule is already fitted, the other half is not fitted), running phenix.dock_in_map before cryo_fit1 is recommended.

Note that these options (phenix.dock_in_map or UCSF Chimera's 'fit in map') do rigid-body fitting only. Therefore, these are useful as global fitting before cryo_fit1. However, when overall orientation is already well fitted, often these are not that needed before cryo_fit1.

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- When fitting protein molecule to a map requires a significant conformational change (for example, half of the molecule is already fitted and the other half is not fitted like

), enforcing stronger `emweight_multiply_by` tends to fit better (obviously).

For example, `phenix.cryo_fit <user.pdb> <user.map> emweight_multiply_by=50` At this high `emweight_multiply_by`, the secondary structures may be broken (slightly or seriously) right after `cryo_fit` running. However, `phenix.model_idealization` which is followed by `phenix.real_space_refine` perfectly restored ideal secondary structures and fitted very well.

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I can't run `phenix.superpose_pdb`s with `cryo_fitted` pdb file

- `phenix.superpose_pdb`s can superimpose only between pdb files that have equally/similarly aligned nucleic names.

This applies to all pdb input files (not only `cryo_fitted` files).

- This applies to all pdb input files (not only `cryo_fitted` files).
- Therefore, consider to align both input files by "`python <PHENIX path>/modules/cryo_fit/steps/9_after_cryo_fit/align_nucleic_acid_name_into_middle/align_nucleic_acid_name_into_middle.py`"

I see "Fatal error: A charge group moved too far between two domain decomposition steps. This usually means that your system is not well equilibrated" at my 8_cryo_fit step.

- Using macOS 10.13.6 helped rather than using Ubuntu 16.04. Maybe macOS has better numerical stability.
- One gromacs expert suggested to try smaller `time_step_for_cryo_fit`.
- However for Doo Nam, simply using macOS solved the problem.
- Less likely, but still a possible reason is that the map weight is too high. Therefore, lowering `emweight_multiply_by` may help.

I see "Fatal error: Incomplete ring in HIS50" at my 8_cryo_fit step

- Doo Nam observed this error when he tried to run `cryo_fit` with `3jch.pdb` that has incomplete Histidine ring atoms.
- He solved with UCSF Chimera

Open pdb file [menu] Select -> Residue -> HIS Tools -> Structure Editing -> Rotamers -> OK (select the most probable rotamer each) File -> Save PDB

- Open pdb file
- [menu]

- Select -> Residue -> HIS
- Tools -> Structure Editing -> Rotamers -> OK
- (select the most probable rotamer each)
- File -> Save PDB
- Pymol and Swissspdb viewer have similar residue fixing functions
- Alternatively, a user may identify residues manually by running
`<phenix-xxxx>/modules/cryo_fit/steps/0_prepare_cryo_fit/count_number_of_atoms_in_each_residue/`

I see "Fatal error: Number of grid cells is zero. Probably the system and box collapsed." at my 8_cryo_fit step.

- step_8 may be full of stepxb_nx.pdb.

Most likely, this means that initial cc is too low for MD simulation. When Doo Nam ran phenix.real_space_refine first, then run real_space_refined atomic model in cryo_fit, it was solved.

Alternatively, improve initial cc by fitting initial atomic model into a map (see "How to improve initial cc?" in this FAQ) Less likely, but still a possible case is when the map weight is too high, lowering emweight_multiply_by may help.

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I see "pdb file cleaning is not done" at my step 1 (Make gro and topology file by regular gromacs)

"I edited out lipids, HEM and other hetero atoms and I verified that they are all gone. However, still my pdb file is not clean enough for gromacs_cryo_fit".

- If a user's lipids, HEM and other hetero atoms are not necessary for now, a user may remove (take out). A user may add those atoms after flexible fit (cryo_fit). Please send the developers (jprajapati@lanl.gov, dgirodat@lanl.gov, doonam.kim@gmail.com) a user's pdb input file only (no map). They will take a look. gromacs version that cryo_fit uses has amber03 forcefield which may not have needed parameters for user's less-usual atoms/residues.
- If a user's lipids, HEM and other hetero atoms are necessary during flexible fitting, Doo Nam recommends to use cryo_fit2 since it supports phenix.eLBOW derived ligand parameters.

I see "Software inconsistency error" at my 8_cryo_fit step

- Doo Nam could not find useful solution for this "Software inconsistency error: Inconsistent DD boundary staggering limits!" message at <http://www.gromacs.org/Documentation/Errors>.
- He observed this message for 2 cases.

When he tried to run `cryo_fit` on Ubuntu 16.0. However, the same data set (map and model) didn't cause any error on his macOS 10.13.6. When he tried to fit a very lowly correlated initial atomistic model to a map (therefore, he observed like step 3600 correlation coefficient: -0.000820). Therefore, he ran `phenix.dock_in_map` first then, provided `placed_model.pdb` as the input model to `cryo_fit`. Then, the error disappeared.

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I see "state.cpt not found, step_8 may be full of stepxb_nx.pdb" at my 8_cryo_fit step

- Literally, steps/8_cryo_fit fold may be full of `stepxb_nx.pdb` files.

Most likely, this means that a `cryo_fit` input pdb file is not yet stable enough for sensitive Gromacs MD simulation (step 8) even after `cryo_fit`'s ample minimization step (e.g. step 4). When Doo Nam ran `phenix.real_space_refine` first, then run `real_space_refined` atomic model in `cryo_fit`, it was solved more than 2 cases. If initial cc is too low, improve initial cc by fitting initial atomic model into a map, either by following "How to improve initial cc?" in this FAQ or adding/copying more atoms. For example, when Doo Nam tried to fit a monomer into a trimer map, this error occurred. However, simply adding dimer atomic models into an input pdb file and running `cryo_fit` again removed the error. Less likely, but still a possible case is when the map weight is too high, lowering `emweight_multiply_by` may help.

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- Less likely, but still a possible case is when the map weight is too high, lowering emweight_multiply_by may help.

I see "step 0 correlation coefficient: nan" at my 8_cryo_fit step.

- This often indicates that the initial atomic model is not placed into a cryo-EM map.

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- If a user observed "Range checking error: Explanation:" error message at 8_cryo_fit step, please refer <I see "step 0 correlation coefficient: nan" and "Range checking error: Explanation:"> part in this FAQ

I see "step 0 correlation coefficient: nan" and "Range checking error: Explanation: During neighbor searching, we assign each particle to a grid based on its coordinates. If your system contains collisions or parameter errors that give particles very high velocities you might end up with some coordinates being +-Infinity or NaN (not-a-number). Obviously, we cannot put these on a grid, so this is usually where we detect those errors. Make sure your system is properly energy-minimized and that the potential energy seems reasonable before trying again." at my 8_cryo_fit step.

- Based on Doo Nam's experiences, this error message may indicate between two cases/scenarios.

Doo Nam observed this error when starting molecule is not stabilized structurally. There are two possible solutions. Run phenix.real_space_refine before cryo_fit. Then, provide real_space_refined molecule into cryo_fit. Then, most problems were solved. When Doo Nam skipped minimization, this error occurred. Since most users minimize the starting structure by default (cryo_fit default), most users do not need care about this. Besides, as of 9/18/2019, the default value of number_of_steps_for_minimization=20000 (which is a lot). It means that the map dimensions need to be larger. Therefore, check map size and solve a problem with VMD/EMAN2/reliion like followings. Like other MD simulations, gromacs need enough map box size to cover the atomic model to run (ziggle and wiggle). Refer Waters seems to be out of the box For example, stuck-out red oxygen atoms outside the right edge of the box are the problem. In order to run any MD simulation (including cryo_fit), a box should be large enough like Make map box size larger (see "How to enlarge map box size?" in this FAQ), and run cryo_fit again. A user can check map box size by VMD. Alternatively, remove sticking out atoms if these are unnecessary, then run cryo_fit again. For protein modeling, Doo Nam would use cryo_fit2 which is not limited by box size requirement.

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I see "The domain decomposition grid has shifted too much" at my 8_cryo_fit step

- Doo Nam observed "Source code file: domdec.c, line: 2633 Fatal error: Step 48520: The domain decomposition grid has shifted too much in the Y-direction around cell 0 1 0", when he used 4 numbers of cores in macOS.
- Inspired by gromacs email list, when he specified a lower `nproc`, the problem was solved.

For example, `phenix.cryo_fit <input.pdb> <input.ccp4> nproc=2`

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I see "The initial cell size (xxx) is smaller than the cell size limit (xxx), change options -dd, -rdd or -rcon, see the log file for details" at my 8_cryo_fit step

- Doo Nam observed this error for 2 cases

When he provided a pdb file that has unspecified/deleted region of a molecule. Even when vmd assisted map box/cell dimension is larger than initial molecule space, this error appeared. When he provided a pdb file that has small number of residues (e.g. 5~16 amino acids). Even when vmd assisted map box/cell dimension is larger than initial molecule space, this error appeared. This error appeared regardless of the existence of CRYST1 and SCALE header.

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I see "Too many LINCS warnings" at my 8_cryo_fit step

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I see "User's provided atomic model had 0.0 cc" in my cryo_fit.overall_log.

- It means that literally user's provided atomic model has no correlation to map to start with.
- See "How to improve initial cc?" in this FAQ, and provide initially aligned atomic model to cryo_fit.

My model is shifted after phenix.model_idealization.

- See 'My model is shifted after phenix.model_idealization' in cryo_fit2_FAQ

When cryo_fit is useful?

- To model dynamic conformational change, cryo_fit is useful.
- To simply fit to cryo-EM map, cryo_fit is useful when required RMSD > 7 Angstrom. When required RMSD <= 7 Angstrom, phenix.real_space_refine tends to do similar fitting with faster speed.

What to do after `cryo_fit`?

- See 'What to do after `cryo_fit`?' in `cryo_fit2_FAQ`

Frequently asked questions about experimental phasing

experimental_phasing.html

Contents

- General
- Experimental phasing (autosol)
- Phaser

General

What can I do if autosol says "this version does not seem big enough"?

Autosol tries to automatically determine the size of solve or resolve, but if your data is very high resolution or a very large unit cell, you can get the message:

```
*****
Sorry, this version does not seem big enough...
(Current value of isizeit is 30)
Unfortunately your computer will only accept a size of 30
with your current settings.
You might try cutting back the resolution
You might try "coarse_grid" to reduce memory
You might try "unlimit" allow full use of memory
*****
```

You cannot get rid of this problem by specifying the resolution with resolution=4.0 because autosol/autobuild/ligandfit uses the resolution cutoff you specify in all calculations, but the high-res data is still carried along.

The easiest solution to this problem is to edit your data file to have lower- resolution data. You can do it like this:

```
phenix.reflection_file_converter huge.sca --sca=big.sca --resolution=4.0
```

or in the GUI, use the reflection file editor.

A second solution is to tell autosol to ignore the high-res data explicitly with one of these commands (on the command line or in the GUI):

```
resolve_command="'resolution 200 4.0'"
solve_command="'resolution 200 4.0'"
resolve_pattern_command="'resolution 200 4.0'"
```

Note the two sets of quotes; both are required for this command-line input. Just one set of quotes is required in the GUI. These commands are applied after all other inputs in resolve/solve/resolve_pattern and therefore all data outside these limits will be ignored.

Why am I not allowed to use a file with FAVG SIGFAVG DANO SIGDANO in autosol or autobuild?

The group of MTZ columns FAVG SIGFAVG DANO SIGDANO is a special one that should normally not be used in Phenix. The reason is that Phenix stores this data as F+ SIGF+ F- SIGF-, but in the conversion process between F+/F- and FAVG/DANO, information is lost. Therefore you should normally supply data files with F+ SIGF+ F- SIGF- (or intensities), or fully merged data (F,SIG) to Phenix routines. As a special case, if you have anomalous data saved as FAVG SIGFAVG DANO SIGDANO you can supply this to autosol, however this requires either that (1) you supply a refinement file with F SIG, or that (2) your data file has a

separate F SIG pair of columns (other than the FAVG SIGFAVG columns that are part of the FAVG/DANO group).

How can I specify a mask for density modification in autosol/autobuild/?

In autobuild you can simply use the command:

```
mask_from_pdb = my_mask_file.pdb
rad_mask_from_pdb = 2
```

where my_mask_file.pdb has atoms in it marking the region to be masked. All points within rad_mask_from_pdb of an atom in my_mask_file.pdb will be considered inside the mask.

If you want to specify a mask in autosol, add this command:

```
resolve_command_list="'model ../../coords.pdb' 'use_model_mask'"
```

where there are " and ' quotes and coords.pdb is the model to use for a mask. Note the "../../" because coords.pdb is in your working directory but when resolve runs the run directory is 2 directories lower, so relative to that directory your coords.pdb is at "../../coords.pdb".

You will know it is working if your resolve_xx.log says:

```
Using model mask calculated from coordinates
```

Note: this command is most appropriate for use with the keyword maps_only=True because phenix.autobuild also uses model=... so that iterative model-building may not work entirely correctly in this case. Two parts that may not function correctly are "build_outside_model" (which will use your model as a mask and not the current one), and evaluate_model (which will evaluate your starting model, not the current model).

What do I do if autobuild says TRIED resolve_extra_huge ...but not OK?

In most cases when you get this error in phenix:

```
TRIED resolve_extra_huge ...but not OK
```

it actually means "your computer does not have enough memory to run resolve_extra_huge". If that is the case then you are kind of stuck unless you have another computer with even more memory+swap space, or you cut back on the resolution of the input data (Note that you have to actually lower the resolution in the input file, not just set "resolution=" because all the data is kept but not used if you just set the resolution).

You can also try the keyword:

```
resolve_command_list="'coarse_grid'"
```

(note 2 sets of quotes)

Sometimes the not OK message can happen if your system and PHENIX are not matching, so that resolve or solve cannot run at all. You can test for this by typing:

```
phenix.resolve
```

and if it loads up (just type QUIT or END or control-C to end it) then it runs, and if it doesn't, there is a system mismatch.

Experimental phasing (autosol)

How can I tell autosol which columns to use from my mtz file?

In the GUI this is handled automatically, and available columns will be loaded into drop-down menus. On the command line, autosol will normally try to guess the appropriate columns of data from an input data file. If

there are several choices, then you can tell autosol which one to use with the command_line keywords labels, peak.labels, infl.labels etc. For example if you have two input datafiles w1 and w2 for a 2-wavelength MAD dataset and you want to select the w1(+) and w1(-) data from the first file and w2(+) and w2(-) from the second, you could use following keywords (see "How do I know what my choices of labels are for my data file" to know what to put in these lines):

```
input_file_list=" w1.mtz w2.mtz"
group_labels_list=" 'w1(+) SIGw1(+) w1(-) SIGw1(-)' 'w2(+) SIGw2(+) w2(-) SIGw2(-)'"
```

Note that all the labels for one set of anomalous data from one file are grouped together in each set of quotes.

You could accomplish the same thing from a parameters file specifying something like:

```
wavelength{
  wavelength_name = peak
  data = w1.mtz
  labels = w1(+) SIGw1(+) w1(-) SIGw1(-)
}
wavelength{
  wavelength_name = infl
  data = w2.mtz
  labels = w2(+) SIGw2(+) w2(-) SIGw2(-)
}
```

How do I know what my choices of labels are for my data file?

You can find out what your choices of labels are by running the command:

```
phenix.autosol show_labels=w1.mtz
```

This will provide a listing of the labels in w1.mtz and suggestions for their use in autosol/autobuild/ligandfit. For example the labels for w1.mtz yields:

```
List of all anomalous datasets in w1.mtz
'w1(+) SIGw1(+) w1(-) SIGw1(-)'

List of all datasets in w1.mtz
'w1(+) SIGw1(+) w1(-) SIGw1(-)'

List of all individual labels in w1.mtz
'w1(+)'
'SIGw1(+)'
'w1(-)'
'SIGw1(-)'

Suggested uses:
labels='w1(+) SIGw1(+) w1(-) SIGw1(-)'
input_labels='w1(+) SIGw1(+) None None None None None None None'
input_refinement_labels='w1(+) SIGw1(+) None'
input_map_labels='w1(+) None None'
```

Why do I get "None of the solve versions worked" in autosol?

If you get this or a similar message for resolve, first have a look at LAST.LOG if it exists in your AutoSol_run_xx_ or AutoBuild_run_xx_ directory. The end of that file may give you a hint as to what was wrong.

The next thing to try is running one of these commands (just kill them with control-C if they do run):

```
phenix.solve
```

or:

```
phenix.resolve
```


If these load up solve or resolve, then they basically work and the problem is probably in the size of your dataset, some formatting issue, or the like.

If they do not run, then the problem is in your system setup. If you are using redhat linux, try changing the option of selinux to selinux=disabled in your /etc/sysconfig/selinux file.

It is also possible that you do not have the application csh installed on your system. If you have Ubuntu linux, csh and tcsh are not included in a normal installation. It is easy to install csh and tcsh under linux and it just takes a minute. On Ubuntu or Debian, you can say:

```
apt-get install tcsh
```

or on Fedora or CentOS and similar distributions:

```
yum install tcsh
```

and that should do it.

How can I do a quick check for iso and ano differences in an MIR dataset?

You can say:

```
phenix.autosol native.data=native.sca deriv.data=deriv.sca
```

and wait a couple minutes until it has scaled the data (once it says "RUNNING HYSS" you are far enough) and then have a look at:

```
AutoSol_run_1_/TEMP0/dataset_1_scale.log
```

which will say near the end:

```
isomorphous differences derivs 1 - native
```

```
Differences by shell:
```

```
shell dmin nobis Fbar R scale SIGNAL NOISE S/N
```

```
1 5.600 1018 285.012 0.287 0.998 105.05 26.73 3.93
2 4.200 1386 324.927 0.216 1.000 84.78 26.76 3.17
3 3.920 542 330.807 0.214 1.002 85.00 28.36 3.00
4 3.710 523 286.487 0.237 1.002 81.31 27.29 2.98
5 3.500 662 282.383 0.235 1.001 75.58 37.12 2.04
6 3.360 518 255.782 0.241 1.003 72.69 27.18 2.67
7 3.220 630 237.778 0.253 1.000 68.87 29.94 2.30
8 3.080 727 208.271 0.255 1.000 61.39 29.19 2.10
9 2.940 897 190.044 0.254 0.999 42.78 42.99 1.00
10 2.800 1067 169.022 0.280 0.999 50.54 33.24 1.52
```

```
Total: 7970 256.096 0.245 1.000 75.29 31.41 2.48
```

Here R is $\langle F_{\text{deriv}} - F_{\text{native}} \rangle / (2 \langle F_{\text{deriv}} + F_{\text{native}} \rangle)$, noise is $\langle \sigma \rangle$, signal is $\sqrt{\langle (F_{\text{deriv}} - F_{\text{native}})^2 \rangle - \langle \sigma^2 \rangle}$, and S/N is the ratio of signal to noise.

I ran autosol to get a partial model that I now want to refine. Which data file should I give as input: the original .sca file from HKL2000, or the file overall_best_refine_data.mtz from autosol?

Always use the MTZ file output by autosol. This contains a new set of R-free flags that have been used to refine the model; starting over with the .sca file will result in a new set of flags being generated, which biases R-free.

Phaser

How can I calculate a log-likelihood gradient (LLG) map in Phaser?

See the maps section of the general FAQ list.

Frequently asked questions (FAQs)

[index.html](#)

Separate FAQ lists are available for experimental phasing [experimental_phasing.html](#), molecular replacement, model building, ligands, refinement. There is also an overview of file formats used by Phenix. You can use the Phenix chatbot to get help on any Phenix program.

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Citing Phenix

How should I cite PHENIX?

If you use PHENIX please cite:

Macromolecular structure determination using X-rays, neutrons and electrons: recent developments in Phenix. D. Liebschner, P. V. Afonine, M. L. Baker, G. Bunkóczi, V. B. Chen, T. I. Croll, B. Hintze, L.-W. Hung, S. Jain, A. J. McCoy, N. W. Moriarty, R. D. Oeffner, B. K. Poon, M. G. Prisant, R. J. Read, J. S. Richardson, D. C. Richardson, M. D. Sammito, O. V. Sobolev, D. H. Stockwell, T. C. Terwilliger, A. G. Urzhumtsev, L. L. Videau, C. J. Williams and P. D. Adams. *Acta Cryst.* (2019). D75, 861-877

Individual programs may have additional citations.

Clarifications

This section is used to clarify subjects.

Can I use Polder for cryo-EM data?

No, Polder is only suitable for crystallographic data

Is there a chatbot for Phenix help?

Yes, there is a free online chatbot that contains all the Phenix documentation and text from all the Phenix tutorial videos. You can ask it any question about how to use Phenix. Note that it requires a Google account (any free account is fine).

Can I get an AI analysis of my results?

You can get an AI analysis of your results using the 'AI Analysis' button on the Results panel of the Phenix GUI for most programs. This analysis will summarize your run and suggest next steps.

What is the normal sequence of steps in Phenix if I have X-ray crystallography data?

If you have X-ray crystallography data, you are normally starting with the sequences of the chains in your molecule, along with X-ray data. Your X-ray data might have an anomalous signal (for example if you have an anomalously-scattering atom like Se and you have collected anomalous data). Your overall strategy is to obtain crystallographic phases and a density map, either by experimental phasing (if you have anomalous data), or by molecular replacement (if you can find related structures or predict the structure). Once you have a map, you will build or rebuild a model, refine it, and find any ligands.

Your first step with X-ray data is analysis of your data with Xtriage. The next step is usually to use PredictAndBuild to obtain a preliminary model and a density map starting with AlphaFold predictions, or if you have data with an anomalous signal, AutoSol and AutoBuild to solve your structure and get a map and model with experimental phasing.

If there are more than one copies of some or all chains in the unique part of your structure (in the asymmetric unit of the crystal), you may want to find the relationships between these chains (non-crystallographic symmetry) with FindNCS.

Once you have a model, with X-ray data normally you will use PhenixRefine to improve the model and automatically carry out comprehensive validation, iterating with Coot or Isolde to rebuild parts of the model that do not fit the density map well.

When you have a very good model with X-ray data, you might use Ligandfit to find the locations of ligands, followed by additional refinement and rebuilding steps.

If you have X-ray data and your model is complete, you will carry out final validation steps.

You can also split these steps in X-ray crystallographic structure determination into sub-steps so that you have more control. Instead of PredictAndBuild, you can use PredictModel to get AlphaFold models for each chain in your molecule, PhaserMR to find the location of each chain, and AutoBuild to rebuild the model. If you have data with an anomalous signal, instead of AutoSol you can run HySS to find the sites of anomalously-scattering atoms, Phaser to calculate phases and a map, and AutoBuild to build a model.

What is the normal sequence of steps in Phenix if I have cryo-EM data?

If you have cryo-EM data, you are normally starting with the sequences of the chains in your molecule and your data, two independent half-maps and a full (averaged) map. Your goal is to optimize the map and create an atomic model that matches your sequences and maps. You can obtain your model by directly building into your map or by docking structures that are similar and adjusting them.

Your first step with cryo-EM data is analysis of your data with Mtriage. You may also want to find the symmetry of your map with MapSymmetry.

The next step with cryo-EM data is usually to use PredictAndBuild to obtain a preliminary model and an improved density map based with AlphaFold predictions.

With cryo-EM data, once you have a model, normally you will use RealSpaceRefine to improve the model and automatically carry out comprehensive validation, iterating with Coot or Isolde to rebuild parts of the model that do not fit the density map well.

When you have a very good model fitting your cryo-EM data, you might use Ligandfit to find the locations of ligands, followed by additional refinement and rebuilding steps. When your model is complete, you will carry out final validation steps.

You can also break down these steps in cryo-EM structure determination into sub-steps so that you have more control. You can use MapSharpening or ResolveCryoEM to improve the clarity of your cryo-EM map. You can predict the structures of the chains in your molecule with PredictModel or you can find a similar structure in the PDB, then you can use DockInMap or EMPlacement to place these chains in your cryo-EM map. You can rebuild a preliminary model or generate a new one from a map using MapToModel.

How do I run MR-SAD in Phenix?

To run MR-SAD in Phenix, you first run MR (molecular replacement) with Phaser, then you run autosol (experimental phasing), including the placed model you obtained with MR with the keyword `partpdb_file=MR.1.pdb` (or whatever your placed model is called).

Is there an AI agent for Phenix?

Yes, there is a Phenix AI agent. You can run it in the GUI or from the command line. It will automatically run many Phenix programs for you.

What is the easiest way to combine two models in Phenix?

You can use `phenix.pdbtools` on the command line to combine two models.

General

What kind of hardware do I need to run Phenix?

A set of loose guidelines is here. To summarize: most consumer-grade systems (including laptops) purchased in the last three years should be sufficient, and at least 2GB of RAM per processor core is recommended. In many cases older hardware is fine and we ourselves continue to run Phenix on (relatively high-end) systems from 2008, but having lots of memory is still essential.

How can I use multiple processors to run a job?

Only Autosol, MR-Rosetta, Autobuild, Ligandfit, phaser.MRage, `phenix.refine`, the structure comparison GUI, and `phenix.find_tls_groups` support runtime configuration of parallel processing. In most cases this is done by adding the "nproc" keyword, for instance:

```
phenix.autobuild data.mtz model.pdb seq.dat nproc=5
```

Equivalent controls are usually displayed in the GUI. Some of these programs (`phenix.refine` is the major exception) also support parallelization over queuing systems; this may require manual input of the queue submission command.

In addition to these options, it is also possible to compile `phenix.refine` and Phaser with the OpenMP library, which automatically parallelizes specific instructions such as the FFT. This requires using the source installer for Phenix, and adding the argument "`--openmp`" to the install command. Because of threading conflicts, OpenMP is not compatible with the Phenix GUI.

How can I run Phenix across multiple computers in our lab?

You will need to install a managed queuing system to handle job submission. Currently we support (to varying degrees) Sun Grid Engine, PBS, LSF, and Condor. Note that this assumes that the available processors are dedicated for cluster use, otherwise the submitted jobs will compete for resources with local processes.

You can use the queuing system in two ways: either to parallelize the execution of subprocesses for various programs listed above, or to submit jobs from the Phenix GUI. However, these two options are not inherently compatible: if you submit a job from the GUI it will usually be unable to submit additional child processes to the queue without further configuration.

Where can I find sample data?

You can find sample data in the directories located in \$PHENIX/examples. Additionally there is sample MR data in \$PHENIX/phaser/tutorial.

In the Phenix GUI, tutorials may automatically be set up as projects. (You must run the specific programs manually, however.)

Can I easily run autosol, autobuild or ligandfit with some sample data?

You can run sample data with with a simple command. To run p9-sad sample data with autosol, you type:

```
phenix.run_example p9-sad
```

This command copies the \$PHENIX/examples/p9-sad directory to your working directory and executes the commands in the file run.sh

What sample data are available to run automatically?

You can see which sample data are set up to run automatically by typing:

```
phenix.run_example --help
```

This command lists all the directories in \$PHENIX/examples/ that have a command file run.sh ready to use. For example:

```
phenix.run_example --help
```

```
PHENIX run_example script. Fri Jul 6 12:07:08 MDT 2007
```

```
Use: phenix.run_example example_name [--all] [--overwrite]
Data will be copied from PHENIX examples into subdirectories
of this working directory
If --all is set then all examples will be run (takes a long time!)
If --overwrite is set then the script will overwrite subdirectories
```

```
List of available examples: 1J4R-ligand a2u-globulin-mr gene-5-mad
p9-build p9-sad
```

Are any of the sample datasets annotated?

The PHENIX tutorials listed on the documentation index will walk you through sample datasets, telling you what to look for in the output files. For example, the Tutorial 1: Solving a structure using SAD data tutorial uses the p9-sad dataset as example. It tells you how to run this example data in autosol and how to interpret the results.

How is non-crystallographic symmetry handled in PHENIX for X-ray structures?

Non-crystallographic symmetry (NCS) in a crystal structure refers to local symmetry that is not part of the crystal lattice itself. For example, within the asymmetric unit of a crystal there might be two copies of a chain related to each other by a two-fold axis that does not align with any axis of the crystal. These two copies are related by local, or non-crystallographic symmetry. Normally the NCS is represented by a set of matrices and translations that relate each copy to a reference copy. This may be written as an NCS file, often with the suffix, '.ncs_spec'.

The treatment of NCS in crystallography depends on the specific program or task:

- In density modification (performed internally by RESOLVE, and used in Autosol, autobuild, phenix.multi_crystal_average, and related programs), NCS relationships (in the form of rotation/translation matrices) are used to average different parts of the map together, which usually improves phase quality. The NCS operators can be derived from heavy-atom positions, the map itself, or

a partial model, but this is normally done automatically (although we recommend specifying the number of NCS-related copies if you know what it should be).

- In model-building (Autobuild), NCS is also used to copy pieces of the partial model to complete related chains.
- In refinement (both in phenix.refine itself, and in the Autosol/Autobuild pipelines), NCS-related chains will be restrained to have similar conformations. By default this will be done in torsion-angle space, which allows for local conformational changes if present, and also incorporates advanced sidechain-fitting features. The NCS groups will be determined automatically based on sequence comparison. Alternately, you may restrain the NCS groups to be globally similar, treating the relationships as rigid. This may work better in a few cases but is generally not recommended, especially since careful manual setup is usually required for optimal results.

Note that several programs may produce multiple forms of output specifying the NCS relationships. Files ending in `.ncs_spec` contain rotation and translation matrices for use in RESOLVE. Files ending in `.phil` contain atom selections for use in phenix.refine. In most cases, however, you can ignore these files, as the NCS relationships will be detected automatically.

What does map symmetry mean in cryo-EM structure determination?

The symmetry of the map in a cryo-EM structure determination is normally referred to as 'map symmetry' or 'map reconstruction symmetry', as it is the symmetry that was applied during the reconstruction of the map.

The symmetry of the map can be referred to by symbols such as C2 or O, and it can be stored as matrices and translations in the same form as non-crystallographic symmetry as stored in crystallography (in files with names like `'map_symmetry.ncs_spec'`).

In some cases the words 'NCS symmetry' may refer to the map symmetry in cryo-EM, but this usage should generally be avoided. Note that the file names with map symmetry do have the word `'ncs_spec'`. This is because the format is identical to that used in crystallography for NCS.

A cryo-EM map can have local symmetry (symmetry of the molecule that is not present in the reconstruction symmetry) as well. This local symmetry is similar to the local symmetry that can be present in crystal structures. Phenix does not have a system to handle local symmetry in cryo-EM structures at present.

How can I get Phenix to use a local PDB mirror instead of fetching entries remotely?

The Phenix GUI, phenix.cif_as_mtz, MRage, MR-Rosetta and several other programs use a common central function to download PDB files. You can substitute your own local PDB mirror by defining environment variables, for instance in bash/sh:

```
export PDB_MIRROR_PDB=/data/pdb_mirror/pdb
export PDB_MIRROR_STRUCTURE_FACTORS=/data/pdb_mirror/structure_factors
```

The structure factors are optional and may be left out to conserve disk space; most applications are only concerned with the model data. In both cases, these directories should contain a series of subdirectories with two-character names corresponding to the middle two characters of PDB IDs, each containing associated PDB files with specific names such as:

```
$PDB_MIRROR_PDB/hb/pdb1hbb.ent.gz
$PDB_MIRROR_STRUCTURE_FACTORS/hb/r1hbbsf.ent.gz
```

If the mirror is out of date and a specific entry cannot be found, the function will revert to downloading from the RCSB PDB site.

Installation

I'm upgrading to a new version of Phenix. How do I uninstall the previous version?

You can simply delete the entire folder, for example `/usr/local/phenix-VERSION` or (on Mac) `/Applications/PHENIX-VERSION`. On Mac, you can drag the folder into the Trash. Each installation is entirely self-contained and all project data is stored elsewhere (for instance in your home directory).

Can multiple installations of Phenix co-exist on the same computer?

Yes, but note that the command names will overlap, so you will either need to source the appropriate `phenix_env` script as needed, or launch programs with the version suffix:

```
phenix.refine-1.8.4-1496
phenix.refine-1.9-1962
```

Graphical interface

When I start up the Phenix GUI for the first time, I get an error message "ImportError: libjpeg.so.62: cannot open shared object file: No such file or directory". How do I fix this?

This error usually occurs on recent versions of Ubuntu, which do not install an essential library by default. You can fix it by running this command:

```
sudo apt-get install libjpeg62
```

How do I find a specific parameter in the GUI?

Open the Settings menu, and select Search all parameters. This can be used to search for any keyword in the parameter name, label, or help text.

Working with reflection files

What reflection file formats does Phenix support?

We recommend using MTZ files where possible but several other formats are also read (including XDS and Scalepack). Programs in Phenix rarely output any format other than MTZ, however. See the overview in the page for file formats for more details.

I have requested that the test set fraction be 0.05 (i.e. 5%), but Phenix only flags 1% of reflections. How do I make it use the number I specify?

By default there is a limit of 2000 on the number of reflections that will be picked for the test set; you don't really need more than this to ensure that R-free is statistically significant and that the maximum likelihood calculations work properly. However, if you want to use a full 5%, you can tell Phenix not to put any upper absolute limit. In the reflection file editor in the GUI, click "More options" in the R-free flags section of the editor, and delete the field labeled "Maximum number of reflections in test set".

Are MTZ files (or R-free flags) output by CCP4 data processing software suitable for use in Phenix?

Yes, absolutely. By default Phenix uses a different convention for labeling the test set when generating new R-free flags, but it will automatically detect the CCP4 convention and adjust as needed.

Phenix generated R-free flags with the test set denoted by 1, but REFMAC expects 0. How can I generate a more compatible MTZ file?

The reflection file editor has an option to export R-free flags to the CCP4 convention. Note however that newer versions of REFMAC will adjust to the Phenix convention automatically, so we recommend updating your CCP4 installation if you encounter this problem.

Should I use the MTZ file output by phenix.refine as input for the next round of refinement?

(Excerpted from the refinement FAQ, since this question is so common.)

The only time this is necessary is when you refined against a dataset that did not include R-free flags, and let phenix.refine generate a new test set. In this case, you should use the file ending in "_data.mtz" for all future rounds of refinement. You do not need to update the input file in each round, as the actual raw data (and R-free flags) are not modified.

Maps

I generated map coefficients in MTZ format using [some Phenix tool]. Why doesn't anything appear when I open it in Coot?

This almost always results from use of the "Auto-open map coefficients" menu item in Coot, which only recognizes several specific MTZ column labels output by programs such as phenix.refine, phenix.maps, Phaser, or REFMAC. (Several other programs in Phenix will now also output the FWT,PHWT columns that are interpreted as a 2mFo-DFc map, although depending on the program these may actually be a density-modified mFo map.) For any other map coefficient labels, for example an anomalous map, you will need to use the next menu item which requires explicit selection of the amplitude and phase labels. You can however modify the Coot preferences to recognize additional map types by editing the file ~/coot.py and adding one or more lines like the following:

```
set_auto_read_column_labels("FEM", "PHIFEM", 0)
```

which would make it possible to auto-open the feature-enhanced maps [../reference/fem.html](http://reference.fem.html). (Thanks to Georg Mlynek for this suggestion.)

Note that in most cases the Phenix GUI will automatically load the appropriate map coefficients for you upon clicking the Coot button in the results tab. Also note that on Macintosh, there are bugs in the X11/XQuartz software which often result in the window to select column labels appearing behind the main Coot window, which is easy to miss.

I have an MTZ file containing map coefficients. How can I obtain a map appropriate for display in PyMOL (or Chimera, etc.)?

In the Phenix GUI, this is almost always automatic: any time a button labeled "Open in PyMOL" is displayed, this will perform the necessary conversion to CCP4 format. In the phenix.refine GUI, there is also a separate button to simply generate the CCP4 maps without opening PyMOL. The map grid spacing, which controls the fineness of the displayed mesh, can be set in the "Graphics" section of the Preferences. By default this is a relatively conservative $d_{\min}/3$ for reasons of speed and disk space. In most cases the map region will cover the atoms in the corresponding PDB file plus a generous buffer.

You may also create CCP4 (or X-PLOR or DSN6) maps manually. In the GUI, the program labeled "FFT map coefficients" in the "Maps" category performs this action; only an MTZ file is required as input, but a PDB file is recommended if you want to ensure that all atoms are covered. Every valid set of map coefficients in the MTZ file will be converted, with the exception of F(model) arrays. On the command line, this program exists as phenix.fft or phenix.mtz2map:

```
phenix.fft refine_001.mtz refine_001.mtz grid_resolution_factor=0.25
```

What is the best way to generate an omit map in Phenix?

There are several options (with some overlap in functionality):

- To remove model bias, the fastest option is to generate a composite omit map for the entire structure. By default this uses a very fast procedure for iteratively de-biasing the phases, but other (slower) options are available too.
- For omitting just a subset of atoms, an older and very popular option is to run phenix.refine with simulated annealing, here specifying an atom selection to be omitted. This also includes an option to harmonically restrain atoms to prevent other parts of the model from drifting into the omitted region.
- The composite omit map tool also has an option to use phenix.refine to de-bias the structure, in addition to the fast method described above. The refinement is much slower, but may be parallelized across multiple CPU cores or a cluster.
- Finally, the most thorough procedure for bias removal is the iterative build omit map option in AutoBuild. This rebuilds the entire model, in addition to running refinement.
- If you just want a simple omit map for a subset of atoms and do not care about model bias, phenix.maps has an option to omit a subset of atoms. This is suitable for quick calculations, but not for generating figures for publication.

Do Phenix maps contain R-free-flagged reflections?

In most cases, yes (except as noted below). For programs such as phenix.refine or phenix.maps, maps may optionally have the flagged reflections excluded, but this is not the default behavior. In the Phenix GUI, the "Refinement" section of the Preferences allows you to set this option globally. Usually this will lead to flagged reflections being substituted by $F(\text{model})$ to prevent Fourier artifacts due to incomplete data; this may slightly increase model bias but it will not lead to biasing R-free if the resulting map coefficients are used for real-space refinement. The program phenix.remove_free_from_map can remove the flagged reflections from an existing MTZ file, and phenix.fft/phenix.mtz2map has an option to perform this when converting to CCP4 or similar format.

Although the previous paragraph applies to output maps, the real-space refinement in phenix.refine always removes flagged reflections from the maps used internally. This is essential to prevent biasing R-free by refining against the real-space map incorporating the flagged reflections.

How do I apply B-factor sharpening or blurring to a map?

Both phenix.refine and phenix.maps allow you to customize the generation of map coefficients and/or CCP4 maps. In either case, there will be a button to enter map options, with several pre-defined map types already entered (at least 2mFo-DFc, mFo-DFc, and anomalous). Each map has a setting to activate map sharpening, and an input field to enter the B-factor to apply. Negative values sharpen the map, positive values blur it.

Note that Coot also allows you to perform these operations interactively (Calculate menu --> Map sharpening...), which may be more helpful than guessing a number.

How do I obtain an anomalous difference map in Phenix?

If you have anomalous data (separate F+ and F- or I+ and I-) and use this as input, both phenix.maps and phenix.refine will automatically output the anomalous difference map in the map coefficients MTZ file. Note that depending on the application, you may need to manually open the MTZ file in Coot to get the anomalous map to appear.

How can I calculate a log-likelihood gradient SAD map using Phaser?

Several programs in Phenix interface with Phaser to calculate an LLG map given a model with existing anomalous (and real) scatterers. This map is more sensitive towards weaker scatterers (such as phosphorous, sulfur, and chloride, or partially-occupied heavy atoms) because it subtracts the contribution of existing atoms. (In this sense it is similar to any other difference map, where the sigma level of the map

around remaining unmodeled atoms becomes increasingly higher as more atoms are added to the model). The first two options are recommended; in both cases you will need to request an additional set of map coefficients where `map_type=LLG`. (This is available as a preset option in the GUI.)

- In `phenix.maps`, if you specify the wavelength parameter, the entire structure will have scattering factors adjusted to model the anomalous scattering expected at the given wavelength. This is relatively fast, but the disadvantage is that the theoretical f'' may be either too high or too low, leading to difference map peaks around existing atoms.
- In `phenix.refine`, if you enable anomalous group refinement and specify the groups to be refined (or manually enter fixed f'' values), the LLG map will reflect the refined values rather than theoretical calculations. You can also set `group_anomalous.find_automatically=True` to identify appropriate groups automatically (based on atomic number).
- If you use the Phaser-EP GUI, there is also an option to output the LLG map coefficients at the end. However, if you turn on anomalous substructure completion, the final LLG map will usually be relatively flat. (Alternately, if you do not use substructure completion, you will be limited to whatever anomalously scattering atoms are present in the input sites file.) Therefore, unless you actually want to perform experimental phasing, when starting from a partial (or mostly complete) model the above methods are more useful.

Finally, for automation purposes, there is a utility for calculating a single set of map coefficients:

```
mmtbx.compute_map_coefficients model.pdb data.mtz wavelength=0.9792 map_type=llg
```

This should produce identical output to `phenix.maps`, but has been streamlined to require minimal input.

I have maps for the same protein(s) in different crystal forms, which I would like to compare. How can I superpose the maps along with the models?

There are two options for this, with slightly different limitations and uses:

- `phenix.superpose_maps`: given two models and two MTZ files containing map coefficients, superposes the models (using user-provided selections if desired), moves them into a P1 box (where all sites $> [0,0,0]$), then transforms the selected maps to follow. Only works on one pair of map coefficients at a time, but allows any type of complex array to be used.
- `structure comparison`: given M models containing N copies of chain X, and M corresponding MTZ files, extracts all copies of chain X, superposes in a P1 box, transforms the maps to follow. Works with any arbitrary number of models or chains, but limited to the most common map types (2mFo-DFc, mFo-DFc, anomalous).

Models

The validation GUI (or the PDB deposition annotator) is complaining about zero-occupancy atoms. Why does this happen, and how can I fix it?

The most common reason for atoms having zero occupancy is heteroatom groups (in this context, any atom labeled as HETATM in the PDB file) being reset to zero occupancy by Phaser. Although this can be avoided by modifying the Phaser settings, it is easy to fix using several different programs:

- In PDBTools, there is an option to reset all selected occupancies to a specified value (usually 1.0). If you do not want any non-unity occupancies it is safe to do this for the entire model, otherwise you will need to enter an atom selection to modify.
- For more manual control, the PDB file editor in the Phenix GUI allows you to change the properties of individual atoms (or residues, chains, etc.).

- In phenix.refine, there are options to modify the starting model in the middle panel of the configuration interface; these are essentially identical to the options in PDBTools. (On the command line, the parameter scope "modify_start_model" contains these options.)

Please note that any time you modify the atomic properties (or add or remove atoms), it is essential to refine the model again, as the previously calculated R-factors and maps will be invalid.

What is the difference between ATOM and HETATM records in a PDB file?

Within Phenix, few distinctions are made between the different atom labels. ATOM generally refers to canonical amino acid residues or nucleic acid bases; HETATM ("heteroatom") covers everything else, including non-canonical amino acids such as selenomethionine (MSE), covalently modified residues, ligands, and solvent. However, this distinction is not enforced (and is frequently ignored) within Phenix, and the choice of label will usually have no effect on program behavior. (The lone exception is Phaser, which by default sets the occupancies of HETATMs to zero unless they are part of a protein polymer.) The PDB annotators will change ATOM to HETATM as needed upon deposition.

Rosetta

(These questions pertain to the Rosetta software for protein structure prediction and design, which is developed and distributed separately from Phenix but is used internally by tools such as MR-Rosetta and phenix.rosetta_refine.)

When I try to build the hybrid Python-enabled Rosetta for running refinement, I get the error below. What does it mean and how can I fix it?

```
[root@localhost ~]# rosetta.build_phenix_interface nproc=8
scons: Reading SConscript files ...
Running versioning script ... Done. (0.0 seconds)
Could not find platform independent libraries <prefix>;
Could not find platform dependent libraries <exec_prefix>;
Consider setting $PYTHONHOME to <prefix>[:<exec_prefix>]
ImportError: No module named site
```

This error happens when the Rosetta build system tries to run Python internally to generate additional files used in compilation. Although we include a complete Python environment with Phenix, Rosetta will probably use the system's Python interpreter (or whatever is defined as the 'python' command). The error message you see means that this interpreter is completely broken, i.e. something is wrong with your system. You can confirm this by running a command such as:

```
python -c "import site"
```

You should consult your local system administrator and/or OS reference manual for assistance fixing the problem.

Other

How can I obtain phases and Hendrickson-Lattman coefficients for my model and data?

There is a program specifically for this purpose; you can find it in the Phenix GUI under "Reflection tools".

Frequently asked questions about ligands

ligands.html

LigandFit wizard

What score indicates an acceptable fit to density?

When using an Fo-Fc difference map as the target, a correlation coefficient (CC) of 0.7 or better usually indicates that the ligand is at least partially correct (with some variation due to differences in data quality or minor inaccuracies in conformation). Values significantly higher than 0.7 almost always indicate an excellent fit to density. A value between 0.6 and 0.7 may not be entirely wrong, but should be treated with suspicion. CC below 0.6 usually means that the ligand is misplaced (which often means that no suitable density could be found). Regardless of the CC, you should always inspect the output model and maps carefully before and after further refinement to evaluate whether the ligand placement is accurate.

Can LigandFit place a peptide for me?

LigandFit works well for single amino acids and may work for small peptides (2-3 residues), but it is not designed to handle larger peptides, which have too many degrees of freedom. You can use AutoBuild to build additional peptides (covered in the model-building FAQs but this will be significantly slower).

What can I do if ligandfit says "this version does not seem big enough"?

Ligandfit tries to automatically determine the size of solve or resolve, but if your data is very high resolution or a very large unit cell, you can get the message:

```
*****
Sorry, this version does not seem big enough...
(Current value of isizeit is 30)
Unfortunately your computer will only accept a size of 30
with your current settings.
You might try cutting back the resolution
You might try "coarse_grid" to reduce memory
You might try "unlimit" allow full use of memory
*****
```

You cannot get rid of this problem by specifying the resolution with resolution=4.0 because ligandfit use the resolution cutoff you specify in all calculations, but the high-res data is still carried along.

The easiest solution to this problem is to edit your data file to have lower- resolution data. You can do it like this:

```
phenix.reflection_file_converter huge.sca --sca=big.sca --resolution=4.0
```

or in the GUI, use the reflection file editor.

A second solution is to tell ligandfit to ignore the high-res data explicitly with one of these commands (on the command line or in the GUI):

```
resolve_command="'resolution 200 4.0'"
solve_command="'resolution 200 4.0'"
resolve_pattern_command="'resolution 200 4.0'"
```

Note the two sets of quotes; both are required for this command-line input. Just one set of quotes is required in the GUI. These commands are applied after all other inputs in resolve/solve/resolve_pattern and therefore all data outside these limits will be ignored.

Restraints generation (eLBOW)

Why is the output geometry incorrect?

With the exception of some especially complex ligands, this problem is almost always caused by the use of a PDB file alone as input (especially when no hydrogens or CONECT records are present). Because the PDB format does not specify the molecular topology unambiguously, it must be guessed based on the coordinates, which is especially error-prone. You should always use a more explicit source of information on the chemistry of your molecule; this does not need to include coordinates (for instance, a SMILES string is appropriate), but it should specify the exact bonding.

I'm using a detailed source of chemical topology, but I'm still getting bad output geometry. How can I fix this?

There are several options:

specify starting geometry for optimization: if you started with a SMILES string, this may provide a more accurate initial estimate than eLBOW is able to guess. use AM1 optimization: AM1 is a semi-empirical quantum-chemistry method included in eLBOW, and is often more robust than the default optimization (at the cost of longer runtime). use Mogul to retrieve ideal geometries from the CCSD: this option searches the large database of small molecule crystal structures for similar geometries (determined at very high resolution). Requires a license for CCSD use and separate installation of the software. optimize with an external quantum chemistry program: eLBOW supports running several popular commercial and academic packages including GAMESS, GAUSSIAN, and Jaguar. (You will need to have these installed separately.) specify explicit output geometry: if you already have accurate geometry from a small-molecule crystal structure of a computational chemistry program and just want Phenix-compatible restraints, you can specify a PDB file containing the exact geometry to use.

- specify starting geometry for optimization: if you started with a SMILES string, this may provide a more accurate initial estimate than eLBOW is able to guess.
- use AM1 optimization: AM1 is a semi-empirical quantum-chemistry method included in eLBOW, and is often more robust than the default optimization (at the cost of longer runtime).
- use Mogul to retrieve ideal geometries from the CCSD: this option searches the large database of small molecule crystal structures for similar geometries (determined at very high resolution). Requires a license for CCSD use and separate installation of the software.
- optimize with an external quantum chemistry program: eLBOW supports running several popular commercial and academic packages including GAMESS, GAUSSIAN, and Jaguar. (You will need to have these installed separately.)
- specify explicit output geometry: if you already have accurate geometry from a small-molecule crystal structure of a computational chemistry program and just want Phenix-compatible restraints, you can specify a PDB file containing the exact geometry to use.

If you encounter especially problematic ligands, we are interested in examining these, as they occasionally expose bugs. You can email us confidentially at help@phenix-online.org.

I know the chemical name of my molecule, but I don't know what residue name has been assigned in the PDB. How do I find this out?

In the eLBOW interface, if you choose to use the PDB Chemical Component Database code to retrieve the molecular structure, you can click a button next to this input field to search the local copy of the database for chemical names matching the search terms.

This will give you a list to choose from (including three-letter codes):

Ligand Identification

What do I expect after a ligand identification run?

On the main GUI status page or towards the end of the command-line output, you will see a list of ligand ranked by scores, and a summary analysis of the results. A typical output is shown below.

Occasionally, if none of the candidate ligands in the library can fit the electron density well enough, especially when the density is much smaller than the ligands to be tested, the program will return an empty list.

OK, I see the output and the list of top ranked ligands. Now what?

The program ranks the fitting results based on objective analysis of several parameters (see document). This usually works well but one should check a few top-ranked ligands in graphics to give the final verdict. The easiest way to do this is to select (highlight) a ligand on the Results page in the GUI and click on one of the display options provided. By default, the fitted ligand, the difference map, and the protein model will be displayed together so you can check if the ligand indeed fits the density well and if it's making good contacts with the protein molecule.

Before displaying the results in graphics, is there a way to know if the program finds anything interesting?

A decent fit usually has a local correlation coefficient (local CC) greater than 0.75. A local CC close to 0.85 usually indicates a near perfect fit. If local CC of a few top-ranked ligands are 0.80 or above, you should be able to get a pretty good idea about what the density might be.

I didn't get an output list. What's the problem?

As said before, this is usually due to that none of the ligands in the library can fit the electron density well enough, especially when the density is much smaller than the ligands to be tested. If your density is the size of just a few atoms, `ligand_identification` is probably not the right tool for you. If this is not the case, try to use a different ligand library may help. See answer to next question.

What ligand library should I use?

The default library is the collection of the top 180 most popular ligands in the PDB. It covers a wide range of molecules in terms of their sizes and chemical nature. Very often, top ranked ligands give you good ideas about the class of ligands and what to try next. Alternatively, one may want to try ligands based on the biological functions of the protein in the crystal. `Ligand_identification` provides several build-in tools to do it on-the-fly. If you know your proteins function in Enzyme Classification (E.C.) terms, E.C. numbers, or Gene Ontology IDs, you can input these information in the run and `Ligand_identification` will compile and use this function-related ligand library for you. For command-line users, you can find a few examples in `Ligand_identification` document.

I think I find a ligand that's close enough, what to do next?

Try to add the ligand into refinement and see what happens.

Frequently asked questions about model-building

model_building.html

General

What can I do if autobuild says "this version does not seem big enough"?

Autobuild tries to automatically determine the size of solve or resolve, but if your data is very high resolution or a very large unit cell, you can get the message:

```
*****
Sorry, this version does not seem big enough...
(Current value of isizeit is 30)
Unfortunately your computer will only accept a size of 30
with your current settings.
You might try cutting back the resolution
You might try "coarse_grid" to reduce memory
You might try "unlimit" allow full use of memory
*****
```

You cannot get rid of this problem by specifying the resolution with resolution=4.0 because autobuild use the resolution cutoff you specify in all calculations, but the high-res data is still carried along.

The easiest solution to this problem is to edit your data file to have lower- resolution data. You can do it like this:

```
phenix.reflection_file_converter huge.sca --sca=big.sca --resolution=4.0
```

or in the GUI, use the reflection file editor.

A second solution is to tell autobuild to ignore the high-res data explicitly with one of these commands (on the command line or in the GUI):

```
resolve_command="'resolution 200 4.0'"
solve_command="'resolution 200 4.0'"
resolve_pattern_command="'resolution 200 4.0'"
```

Note the two sets of quotes; both are required for this command-line input. Just one set of quotes is required in the GUI. These commands are applied after all other inputs in resolve/solve/resolve_pattern and therefore all data outside these limits will be ignored.

Why am I not allowed to use a file with FAVG SIGFAVG DANO SIGDANO in autobuild?

The group of MTZ columns FAVG SIGFAVG DANO SIGDANO is a special one that should normally not be used in Phenix. The reason is that Phenix stores this data as F+ SIGF+ F- SIGF-, but in the conversion process between F+/F- and FAVG/DANO, information is lost. Therefore you should normally supply data files with F+ SIGF+ F- SIGF- (or intensities), or fully merged data (F,SIG) to Phenix routines. As a special case, if you have anomalous data saved as FAVG SIGFAVG DANO SIGDANO you can supply this to autosol, however this requires either that (1) you supply a refinement file with F SIG, or that (2) your data file has a separate F SIG pair of columns (other than the FAVG SIGFAVG columns that are part of the FAVG/DANO group).

How can I specify a mask for density modification in autosol/autobuild/?

In autobuild you can simply use the command:

```
mask_from_pdb = my_mask_file.pdb
rad_mask_from_pdb = 2
```


where my_mask_file.pdb has atoms in it marking the region to be masked. All points within rad_mask_from_pdb of an atom in my_mask_file.pdb will be considered inside the mask.

If you want to specify a mask in autosol, add this command:

```
resolve_command_list="'model ../../coords.pdb' 'use_model_mask'"
```

where there are " and ' quotes and coords.pdb is the model to use for a mask. Note the "../../" because coords.pdb is in your working directory but when resolve runs the run directory is 2 directories lower, so relative to that directory your coords.pdb is at "../../coords.pdb".

You will know it is working if your resolve_xx.log says:

```
Using model mask calculated from coordinates
```

Note: this command is most appropriate for use with the keyword maps_only=True because phenix.autobuild also uses model=... so that iterative model-building may not work entirely correctly in this case. Two parts that may not function correctly are "build_outside_model" (which will use your model as a mask and not the current one), and evaluate_model (which will evaluate your starting model, not the current model).

What do I do if autobuild says TRIED resolve_extra_huge ...but not OK?

In most cases when you get this error in phenix:

```
TRIED resolve_extra_huge ...but not OK
```

it actually means "your computer does not have enough memory to run resolve_extra_huge". If that is the case then you are kind of stuck unless you have another computer with even more memory+swap space, or you cut back on the resolution of the input data (Note that you have to actually lower the resolution in the input file, not just set "resolution=" because all the data is kept but not used if you just set the resolution).

You can also try the keyword:

```
resolve_command_list="'coarse_grid'"
```

(note 2 sets of quotes)

Sometimes the not OK message can happen if your system and PHENIX are not matching, so that resolve or solve cannot run at all. You can test for this by typing:

```
phenix.resolve
```

and if it loads up (just type QUIT or END or control-C to end it) then it runs, and if it doesn't, there is a system mismatch.

Model-building (AutoBuild, etc.)

How do I run AutoBuild on a cluster?

Phenix.autobuild is set up so that you can specify the number of processors (nproc) and the number of batches (nbatch). Additionally you will want to set two more parameters:

```
run_command="command you use to submit a job to your system"
background=False # probably false if this is a cluster, true if this is a multiprocessor machine
```

If you have a queueing system with 20 nodes, then you probably submit jobs with something like:

```
"qsub -someflags myjob.sh" # where someflags are whatever flags you use
```

(or just "qsub myjob.sh" if no flags)

Then you might use:

```
run_command="qsub -someflags" background=False nproc=20 nbatch=20
```

If you have a 20-processor machine instead, then you might say:

```
run_command=sh background=True nproc=20 nbatch=20
```

so that it would run your jobs with sh on your machine, and run them all in the background (i.e., all at one time).

Why does autobuild say it is doing 2 rebuild cycles but I specified one?

The AutoBuild wizard adds a cycle just before the rebuild cycles in which nothing happens except refinement and grouping of models from any previous build cycles.

What is the difference between overall_best.pdb and cycle_best_1.pdb in autobuild?

Autobuild saves the best model (and map coefficient file, etc) for each build cycle nn as cycle_best_nn.pdb. Also the Wizard copies the current overall best model to overall_best.pdb. In this way you can always pull the overall_best.pdb file and you will have the current best model. If you wait until the end of the run you will get a summary that lists the files corresponding to the best model. These will have the same contents as the overall_best files.

How do I tell autobuild to use phenix.refine maps instead of density-modified maps for model-building?

To use the phenix.refine maps instead of density-modified maps, use the keyword two_fofc_in_rebuild=True.

How do I include a twin law for refinement in autobuild?

You can include the twin law in autobuild for refinement with the keyword refine_eff_file=refinement_params.eff, where refinement_params.eff says something like:

```
refinement {  
  twinning {  
    twin_law = "-k, -h, -l"  
  }  
}
```

(You can get the twin law "-k, -h, -l" from phenix.xtriage.)

AutoBuild seems to be taking a long time. What is the usual time for a run?

For typical structures, autobuild runs can take from 30 minutes to several days using a single processor. You can speed up your jobs by using several processors with a command such as "nproc=4". for autobuild you can speed up by up to a factor of 5 in this way. You can also speed up rebuild_in_place autobuild jobs (where your model is being adjusted, not built from scratch) by specifying fewer cycles: "n_cycle_rebuild_max=1" will use 1 cycle of rebuilding instead of the usual 5. Often that is plenty.

Why does autobuild bomb and say "Corrupt gradient calculations"?

If an atom is placed very near a special position then sometimes refinement will fail and an error message starting with "Corrupt gradient calculations" is printed out. If the starting PDB file has the atom near a special position, then the best thing to do is move it away from the special position. If AutoBuild builds a model that has this problem, then it may be easier to rerun the job, specifying "ignore_errors_in_subprocess=True" which should allow it to continue past this error (by simply ignoring that refinement step). You can also try setting correct_special_position_tolerance=0 (to turn off the check) or correct_special_position_tolerance=5 (to check over a wider range of distances from the special position; default=1).

Why does autobuild bomb and say it cannot find a TEMP file?

By default autobuild splits jobs into one or more parts (determined by the parameter "nbatch") and runs them as sub-processes. These may run sequentially or in parallel, depending on the value of the parameter "nproc". In some cases the running of sub-processes can lead to timing errors in which a file is not written fully before it is to be read by the next process. This appears more often when jobs are run on nfs-mounted disks than on a local disk. If this occurs, a solution is to set the parameter "nbatch=1" so that the jobs not be run as sub-processes. You can also specify "number_of_parallel_models=1" which will do much the same thing. Note that changing the value of "nbatch" will normally change the results of running the Wizard. (Changing the value of "nproc" does not change the results, it changes only how many jobs are run at once.)

Is there anyway to get phenix.autobuild to NOT delete multiple conformers when doing a SA-omit map?

At present, if you put you multiple conformations in for the protein autobuild will take only conformation 1 and it will ignore the others.

As a work-around, you can try this: call all the protein a "ligand" and put it in this way (you need to give it one complete residue in the model as "one_residue.pdb" (or any part of the model that has just one conformation):

```
phenix.autobuild data=data.mtz \  
model=one_residue.pdb \  
input_lig_file_list=model.pdb \  
composite_omit_type=sa_omit
```

Autobuild treats ligands as a fixed structure during model building and in omit maps, only adjusted during refinement, which is what you want in this case.

Why does autobuild just stop after a few seconds?

When you run autobuild from the command line it writes the output to a file and says something like:

```
Sending output to AutoBuild_run_3_/AutoBuild_run_3_1.log
```

Usually if something goes wrong with the inputs then it will give you an error message right on the screen. However a few types of errors are only written to the log file, so if autobuild just stops after a few seconds, have a look at this log file and it should have an error message at the end of the file.

What is an R-free flags mismatch?

When you run autoBuild or phenix.refine you may get this error message or a similar one:

```
*****  
Failed to carry out AutoBuild_build_cycle:  
Please resolve the R-free flags mismatch.  
*****
```

Phenix.refine keeps track of which reflections are used as the test set (i.e., not used in refinement but only in estimation of overall parameters). The test set identity is saved as a hex-digest and written to the output PDB file produced by phenix.refine as a REMARK record:

```
REMARK r_free_flags.md5.hexdigest 41aea2bced48fbb0fde5c04c7b6fb64
```

Then when phenix.refine reads a PDB file and a set of data, it checks to make sure that the same test set is about to be used in refinement as it was in the previous refinement of this model. If it does not, you get the error message about an R-free flags mismatch.

Sometimes the R-free flags mismatch error is telling you something important: you need to make sure that the same test set is used throughout refinement. In this case, you might need to change the data file you are using to match the one previously used with this PDB file. Alternatively you might need to start your refinement over with the desired data and test set.

Other times the warning is not applicable. If you have two datasets with the same test set, but one dataset has one extra reflection that contains no data, only indices, then the two datasets will have different hex digests even though they are for all practical purposes equivalent. In this case you would want to ignore the hex-digest warning.

If you get an R-free flags mismatch error, you can tell autobuild to ignore the warning with:

```
skip_hexdigest=True
```

and you can tell phenix.refine to ignore it with:

```
refinement.input.r_free_flags.ignore_pdb_hexdigest=True
```

You can also simply delete the REMARK record from your PDB file if you wish to ignore the hex-digest warnings.

Can I use the autobuild wizard at low resolution?

The standard building with AutoBuild does not work very well at resolutions below about 3-3.2 Å. In particular, the wizard tends to build strands into helical regions at low resolution. However you can specify "helices_strands_only=True" and the wizard will just build regions that are helical or beta-sheet, using a completely different algorithm. This is much quicker than standard building but much less complete as well.

My autobuild composite OMIT job crashed because my computer crashed. Can I go on without redoing all the work that has been done?

Yes, but it involves several steps:

- Run your job again in a separate directory, specifying omit_box_start and omit_box_end to define which omit regions you want to still run. You can figure out how many there should be total from your log file which will say something like: Running separate sub-processes for 12 omit regions. Then as they are running the log file will say what ones are being worked on.

Run your job again in a separate directory, specifying omit_box_start and omit_box_end to define which omit regions you want to still run. You can figure out how many there should be total from your log file which will say something like:

```
Running separate sub-processes for 12 omit regions.
```

Then as they are running the log file will say what ones are being worked on.

- You will now have 2 OMIT/ subdirectories, one from each of your AutoBuild runs.

You will now have 2 OMIT/ subdirectories, one from each of your AutoBuild runs.

- Put all the files together in one directory, and then run an edited version of the script below to combine them:

```
#!/bin/sh phenix.resolve <<EOD hklin exptl_fobs_phases_freeR_flags.mtz labin FP=FP  
SIGFP=SIGFP solvent_content 0.6 no_build combine_map  
overall_best_denmod_map_coeffs.mtz_OMIT_REGION_1 combine_map  
overall_best_denmod_map_coeffs.mtz_OMIT_REGION_2 combine_map  
overall_best_denmod_map_coeffs.mtz_OMIT_REGION_3 combine_map  
overall_best_denmod_map_coeffs.mtz_OMIT_REGION_4 combine_map  
overall_best_denmod_map_coeffs.mtz_OMIT_REGION_5 combine_map  
overall_best_denmod_map_coeffs.mtz_OMIT_REGION_6 combine_map  
overall_best_denmod_map_coeffs.mtz_OMIT_REGION_7 combine_map  
overall_best_denmod_map_coeffs.mtz_OMIT_REGION_8 combine_map  
overall_best_denmod_map_coeffs.mtz_OMIT_REGION_9 combine_map  
overall_best_denmod_map_coeffs.mtz_OMIT_REGION_10 combine_map
```

```
overall_best_denmod_map_coeffs.mtz_OMIT_REGION_11 combine_map
overall_best_denmod_map_coeffs.mtz_OMIT_REGION_12 omit EOD
```

Put all the files together in one directory, and then run an edited version of the script below to combine them:

```
#!/bin/sh
phenix.resolve &&&EOD
hklin exptl_fobs_phases_freeR_flags.mtz
labin FP=FP SIGFP=SIGFP
solvent_content 0.6
no_build
combine_map overall_best_denmod_map_coeffs.mtz_OMIT_REGION_1
combine_map overall_best_denmod_map_coeffs.mtz_OMIT_REGION_2
combine_map overall_best_denmod_map_coeffs.mtz_OMIT_REGION_3
combine_map overall_best_denmod_map_coeffs.mtz_OMIT_REGION_4
combine_map overall_best_denmod_map_coeffs.mtz_OMIT_REGION_5
combine_map overall_best_denmod_map_coeffs.mtz_OMIT_REGION_6
combine_map overall_best_denmod_map_coeffs.mtz_OMIT_REGION_7
combine_map overall_best_denmod_map_coeffs.mtz_OMIT_REGION_8
combine_map overall_best_denmod_map_coeffs.mtz_OMIT_REGION_9
combine_map overall_best_denmod_map_coeffs.mtz_OMIT_REGION_10
combine_map overall_best_denmod_map_coeffs.mtz_OMIT_REGION_11
combine_map overall_best_denmod_map_coeffs.mtz_OMIT_REGION_12
omit
EOD
```

You will want to edit this to match the number of OMIT regions in your case.

Does the RESOLVE database of density distributions contain RNA/protein examples?

The RESOLVE database doesn't have RNA+protein in it, nor does it have low-resolution histograms, but you can create a new entry very easily. Here is how:

- Find a PDB structure that has the characteristics that you want "refine_1.pdb"

Find a PDB structure that has the characteristics that you want "refine_1.pdb"

- Calculate a map with this model at the resolution you are interested in: phenix.fmodel high_resolution=5 refine_1.pdb

Calculate a map with this model at the resolution you are interested in:

```
phenix.fmodel high_resolution=5 refine_1.pdb
```

- Generate histograms with phenix.resolve. Here is a script:

```
#!/bin/sh phenix.resolve <<EOD hklin refine_1.pdb.mtz labin FP=FMODEL PHIB=PHIFMODEL get_histograms no_build solvent_content 0.5 database 5 mask_cycles 1 minor_cycles 1 EOD
```

Generate histograms with phenix.resolve. Here is a script:

```
#!/bin/sh
phenix.resolve &&&EOD
hklin refine_1.pdb.mtz
labin FP=FMODEL PHIB=PHIFMODEL
get_histograms
no_build
solvent_content 0.5
database 5
mask_cycles 1
minor_cycles 1
EOD
```

-Now the file hist_values.dat will have your histograms:

```
5.002693 32.71021 ! resolution Boverall
1 ! 1=protein 2 = solvent
0.10198E-01 1.8145 0.41525E-01 ! a1 a2 a3
```

```
0.14425E-01 0.46920 0.77521 ! a4 a5 a6
0.23653E-06 0.34718E-08 0.0000 ! a7 a8 a9
```

```
2 ! 1=protein 2 = solvent
0.27101E-01 6.4460 -0.61802 ! a1 a2 a3
0.12788E-01 0.55421 -0.39797E-02 ! a4 a5 a6
0.0000 0.0000 0.0000 ! a7 a8 a9
```

- Paste the contents of hist_values.dat at the end of \$PHENIX/solve_resolve/ext_ref_files/segments/rho.list . NOTE: you need one blank line between sections, and an extra 2 blank lines at the very end of the file...otherwise resolve will give a bad error message.

Paste the contents of hist_values.dat at the end of \$PHENIX/solve_resolve/ext_ref_files/segments/rho.list .
NOTE: you need one blank line between sections, and an extra 2 blank lines at the very end of the file...otherwise resolve will give a bad error message.

- Now when you run phenix.resolve...say "database 7" and it will use your new histograms. It will write a message in the log file like this: Histogram DB entry # 7 ("5 14.27721 ! resol")

Now when you run phenix.resolve...say "database 7" and it will use your new histograms. It will write a message in the log file like this:

```
Histogram DB entry # 7 ("5 14.27721 ! resol")
```

which should match what you pasted in to the rho.list file...so you know it took your histograms.

If I run autobuild with after_autosol=True, how do I know which run of autosol it will use?

Autobuild will look through all the autosol runs and choose the solution with the highest final score, and use that one. You can see this near the beginning of the autobuild run:

```
Appending solution 4060.75360229 1 75.3602294036
exptl_fobs_phases_freer_flags_1.mtz solve_1.mtz
Appending solution 59.3469818876 2 59.3469818876 None solve_2.mtz
Best solution 4060.75360229 1 75.3602294036
exptl_fobs_phases_freer_flags_1.mtz solve_1.mtz AutoSol_run_2_
```

In this case it took run 2 with the solution solve_1.mtz with score of 4060.7 over the solution solve_2.mtz with score of 59.

If you want to choose a different autosol solution, then you will need to explicitly tell autobuild all the files that you want to use:

```
phenix.autobuild data=AutoSol_run_5/exptl_fobs_freer_flags_3.mtz \
map_file=AutoSol_run_5/resolve_3.mtz \
seq_file=my_seq_file.seq
```

Notes:

- Use resolve_xx.mtz as a map file, never as a "data" file. It contains coefficients for a density-modified map
- It is recommended not to include the model from autosol in your autobuild runs. Autobuild is a lot better at building a model.
- The data file for autobuild should be AutoSol_run_5/exptl_fobs_freer_flags_3.mtz in this case; note that the "3" here matches the "3" in resolve_3.mtz and is for solution #3 of run 5.
- To see what files to use here, see the file AutoBuild_run_5/AutoBuild_summary.dat which lists the solutions for run 5, and all the files that go with each solution.

Is there a way to use autobuild to combine a set of models created by multi-start simulated annealing?

You can do this in two ways. Both involve the keyword:

```
consider_main_chain_list="pdb1.pdb pdb2.pdb pdb3.pdb"
```

which lets you suggest a set of models to autobuild to consider in model-building.

You can use this with `rebuild_in_place` (all your models should have the same atoms, just with different coordinates):

```
phenix.autobuild data.mtz map_file=map.mtz seq_file= seq.dat \  
model=coords1.pdb rebuild_in_place=True merge_models=true \  
consider_main_chain_list=" coords2.pdb coords3.pdb" \  
number_of_parallel_models=1 n_cycle_rebuild_max=1
```

You can also use it with `rebuild_in_place=False` (any fragments or models are ok):

```
phenix.autobuild data.mtz map_file=map.mtz seq_file= seq.dat \  
model=coords1.pdb rebuild_in_place=False \  
consider_main_chain_list=" coords2.pdb coords3.pdb" \  
number_of_parallel_models=1 n_cycle_rebuild_max=1
```

I ran AutoSol to get a partial model that I now want to refine. Which data file should I give as input: the original .sca file from HKL2000, or the file `overall_best_refine_data.mtz` from AutoSol?

Always use the MTZ file output by AutoSol. This contains a new set of R-free flags that have been used to refine the model; starting over with the .sca file will result in a new set of flags being generated, which biases R-free.

I have a partial model that I used as input for AutoBuild. The output model contains the same atoms and hasn't added anything. Why doesn't it build additional residues?

If the input model matches the sequence (above 50% identity cutoff) and `rebuild_in_place=Auto` (the default), AutoBuild will only rebuild existing atoms without adding or removing anything. Setting `rebuild_in_place=False` will instead tell AutoBuild to generate an entirely new model (although it will also use good parts of the input model in the process), but this risks throwing out some existing atoms. Alternately, see the answer to the next question.

I have a partial model - how do I tell AutoBuild to build new atoms without changing the existing ones?

Specify the partial model as "ligands" (`input_lig_file_list`), and it will be included in refinement but otherwise left alone. This may not extend existing chains, however (only build new chains).

Maps

How can I include high-resolution data and phase extend my map?

You can do this in autobuild with:

```
phenix.autobuild data=data.mtz hires_file=high_res_data.mtz maps_only=True
```

There are many variations on using `maps_only=True` as a way to run density modification. You can also specify a model with `model=mymodel.pdb` and the model information will be used in density modification. If you have a model you can also specify `ps_in_rebuild=True` to get a prime-and-switch map.

When should I use multi-crystal averaging?

Multi-crystal averaging is going to be useful only if the crystals are completely different or the amplitudes are nearly uncorrelated. In cases where there are only small changes the averaging procedure has almost nothing different in the two structures to work with and it won't do much. Another way to say this is that multi-crystal averaging works because two or more very different ways of sampling the Fourier transform of

the molecule are occurring, and each must be consistent with the corresponding measured data. If the molecules are nearly the same and the measured data are nearly the same in all cases, then there are few constraints on the phases.

Yes, experimental phases can be included in multi-crystal averaging, just as for NCS averaging. And yes, experimental phases are most helpful.

If some regions are different in the different crystals, then the masking procedure needs to be adjusted to exclude the variable regions from the averaging process.

Can I make density modified phase combination (partial model phases and experimental phases) in PHENIX?

Yes, you get these if you use:

```
phenix.autobuild model=partial_model.pdb data=exptl_phases_hl_etc.mtz  
rebuild_in_place=False seq_file=seq.dat
```

The model is used to generate phases by a variation on statistical density modification. These phases are then combined with the experimental phases and then the combined phases are density modified. Then the result is density modified including the model. So the file image.mtz is exptl phases + model phases, and image_only_dm.mtz is image.mtz, density modified. Then resolve_work.mtz is image_only_dm.mtz, density modified further using the model as a target for density modification along with histograms, solvent flattening, ncs, etc.

What are my options for OMIT maps if I have 4 fold NCS axis?

Using the keyword omit_box_pdb is a good way of omitting a single small region, or a series of small regions, one at a time. If you want to get a complete sa_omit map or many regions, then skip the omit_box_pdb command and let autobuild make a composite omit map covering the whole a.u.. Use the omit_box_pdb to define a single region that you want omitted (such as a few residues or a loop...)

If you have ncs, you cannot conveniently delete all the copies at once with omit_box_pdb. You can delete the 4 copies one at a time by specifying a list of omit regions however. To omit a list of regions, do it like this:

```
omit_res_start_list="100 500" omit_res_end_list="200 600"  
omit_chain_list="L M"
```

to omit chain L residues 100-200 and then separately chain M residues 500-600.

It shouldn't matter much if you turn off ncs while doing an omit map because the ncs copy won't be used in density modification during the process. However NCS will be used to restrain any coordinates.

Why does AutoSol or AutoBuild say my R/Rfree is 999?

If AutoBuild is used to carry out density modification you may see a line like: Cycle 2 R/Rfree=999.90/999.90 Built=0 Placed=0. This just means that cycle 2 did not involve any refinement in maps_only mode. It does not say anything about your model.

Frequently asked questions about molecular replacement

mr.html

Contents

- General issues
- Refinement after MR
- Phaser-MR GUI
- MRage
- MR-Rosetta
- Reference

General issues

How do I know if Phaser has solved my structure?

This is answered in detail in the section "Has Phaser solved it?" in the main Phaser manual, but it can be summarized as follows: the final translation function Z score (TFZ) should be above 8, and the log-likelihood gain (LLG) should be positive and as high as possible. Of course, as with experimental phasing, the ability to autobuild most of the model is the single best measure of a correct solution; alternately, running a single round of refinement should result in an R-free below 0.50. Note that if the R-free is above 0.5 after refinement, this often indicates that the solution is actually incorrect - especially when combined with sub-standard TFZ and/or LLG values. Just because Phaser finds a solution, does not mean that it has correctly solved the structure!

Why doesn't Phaser report an R-factor?

The R-factor is far less sensitive than the TFZ and LLG scores, especially when searching for remote homologs or a small part of the overall scattering mass. Phaser does record R-factors for the rigid-body refinement step towards the end of a run, which can be found in the logfile. However, unless these values are relatively low (< 40%), they are not reliable indicators of solution quality, which is why only the Z-scores and LLG are reported in the GUI.

What if I don't get a good solution?

This is also covered in detail in the main Phaser manual, in the section "What to do in difficult cases".

What resolution cutoff should I use?

In short, in most cases you should let Phaser choose the resolution cutoffs. Phaser uses the assumed accuracy of the model (expressed directly as RMSD or indirectly as sequence identity), the size of the model, and both the resolution and number of diffraction observations to determine the signal that will be obtained for each model as a function of resolution. This is used both to determine a resolution cutoff that should be sufficient for a successful MR search, and an optimal search order for the models. The effects of bulk solvent at low resolution are taken into account internally, so you do not need to specify a low-resolution cutoff either.

Should I use the output MTZ file from Phaser for refinement?

No, it is always best to use the original data file for refinement. The MTZ file from Phaser contains map coefficients, but in most cases you will find that better maps can be obtained by immediately running a round of refinement or automated re-building.

How are the ASU contents and search ensembles related?

Technically, they aren't connected. The ensembles are used in the actual MR search; the ASU contents only tells Phaser what to expect for the overall scattering mass of the asymmetric unit (an important factor in the maximum likelihood scoring). Although you may only specify one sequence per component, you can instead provide a molecular weight for a collection of chains. You may also split a model into smaller ensembles (for instance, different domains) while keeping the ASU contents the same, or use a multi-chain search model while specifying a sequence file for each chain.

How can I make Phaser go faster?

Phaser is under continuous development, with a lot of emphasis being placed on speed, so anything we could advise here is probably already being done automatically! Because Phaser now automatically chooses the lowest resolution expected to give a clear solution, changing the resolution limits is not advisable and may even lengthen the run times. It's a good idea to set up a Phaser run to find all copies of all components in one job, because this allows Phaser to choose the optimal search order and to amalgamate solutions for multiple copies found in one search.

Can PHENIX do MRSAD?

Yes, PHENIX can run MRSAD (molecular replacement, combined with SAD phases) by using molecular replacement information in the process of determining the anomalous scatterer substructure. However, this requires separate steps for carrying out the MR and finding the substructure. There are two simple ways to do this; both are described in the AutoSol documentation.

I am using a dimer as my search model. Why am I getting a warning that the unit cell is too full?

Remember, the contents of the asymmetric unit is specified independently of the model. If you have supplied a single sequence to define the ASU contents, check to make sure that you have specified that the contents of the asymmetric unit include two copies.

I want to create an ensemble of models. Should I run Sculptor first and then Ensembler, or the other way round?

Unfortunately, neither way is optimal. When Sculptor is run first, it removes residues from the model, and this may impact on Ensembler's ability to do an optimal superposition. If the default SSM matching is used, the fragmentation of secondary structure elements (prominent when models are distant) may result in incorrect matching, although this is rare. When Ensembler is run first, and trimming is switched on, the resulting main-chain trace can contain many short fragments, which will be discarded by Sculptor since these cannot be reliably matched to the alignment sequence, and the completeness of the resulting model is further decreased. The best compromise is probably to run Sculptor first, so that the sequence-to-chain alignment could be performed optimally. In common cases, this does not impair superposition drastically, since there are still many sites to use. In case SSM alignment fails, other matching algorithms (e.g. residue sequence numbers) can be used.

Refinement after MR

What should I run after Phaser finds a good solution?

We usually recommend one (or both) of these two options (and see the next question for advice on more difficult cases):

- If the search model is the same protein (or close enough) and is not expected to have large conformational changes, run phenix.refine using the default strategy for at least 8-10 macrocycles, but with options adjusted as appropriate for the resolution of the data: for instance, adding NCS restraints at

low resolution if multiple copies of a chain are present, or adding the solvent update at moderate-to-high resolution.

- Run AutoBuild; if the search model is of a significantly different protein, turn off "rebuild-in-place" mode so it will build an entirely new model.

I ran Phaser, it found a solution, then I ran phenix.refine, and the R-free is stuck above 0.4 (but below 0.5). How do I fix this?

This is an open-ended question, and there are multiple explanations and solutions possible. We usually recommend a standard series of checks:

- Run Xtriage and check your data for any pathologies that may make refinement difficult (such as twinning, incomplete data, or translational NCS).
- It is very important that the search model already match the sequence of the target protein. If you run Phaser with a homologous structure as input, you should first run Sculptor to edit the search model as needed. (This is not the same as homology modeling - it simply removes inappropriate atoms, renames and renumbers residues, and adjusts B-factors.) Refinement will not correct any residual problems if you skip this step.
- Check whether the MR solution is truly complete - this means that you have placed enough copies of the search model, and that the search model already covers most or all of the target. Gaps in crystal packing or unusually large blobs of difference density may be indicative of missing scatterers. At low resolution the difference density will be weaker, but you can still inspect the lattice. If you are missing significant parts of the model, this will limit how much lower R-factors can get.
- Make sure you are running enough cycles of refinement. Sometimes it will take a long time to converge; the default three cycles is the minimum recommended.

If all of the both indicate that model and data are reasonably sensible, it is likely that there are differences/model errors that are beyond the radius of convergence of refinement. Fortunately, this is usually easy to recognize in the difference maps even at low resolution; for instance, a representative test case at 4.0Å (described in DiMaio et al. 2013):

Here we can see very clearly that the model is simply misfit (although the majority of the protein, mostly outside the field of view, is more or less correct). In this case, there are several tools you can try, many of which were specifically designed to cope with this class of problems:

- If you have moderate-to-high-resolution data (at least 3.0Å), you can try building an entirely new model in AutoBuild. This is slower than refinement, but it has a very wide radius of convergence.
- The model morphing procedure may be helpful; this is available both as an option in AutoBuild, and a standalone program.
- The hybrid Rosetta-Phenix refinement was also designed to deal with cases like this.
- DEN refinement was also designed to deal with structures that need more aggressive improvement, and has proven useful in many similar cases.
- In extreme cases, MR-Rosetta may be tried; this is capable of improving very poor MR solutions, but is significantly more computationally expensive than the other methods.
- Sometimes just adding simulated annealing in phenix.refine will be enough (although for the example pictured above, it was not).

Additionally, if multiple structures of the same protein are available for use as search models, we recommend checking whether any of these is closer to the target than the current search model.

Phaser-MR GUI

Do I actually have to include multiple models in my ensembles?

No, we use "ensemble" in this context to refer to any search model; in many cases a single model is adequate to solve the structure (and these may be much easier to work with).

What is the difference between "high resolution" and "high resolution for refinement"?

The latter is applied during the "refinement and phasing" step only; for rotation and translation searches, the first parameter is used.

How do I search for multiple unique ensembles?

The "Search procedure" tab allows you to set up this procedure. To search for multiple independent ensembles representing different entities in a complex ("a AND b AND c"), click the "Add search" button below the tab, and for each search, check the ensemble(s) to be used at that step. The searches will be performed in the order they appear in the GUI. Note that this is not the same as checking multiple ensembles for a single search, as described in the next question.

How do I tell Phaser to pick the best solution from among several similar ensembles?

To try multiple alternative ensembles (corresponding to the same actual entity in the crystal) at once ("a OR b OR c"), check the box next to each ensemble you want Phaser to try for a single search; only one of these will actually be placed in the output model. (You can, of course, specify any combination of searches and ensembles, for instance "(a OR b) AND c", which would be two separate search instructions.)

I want to run a new search in Phaser starting from the results of a previous job. How do I specify this?

If the previous job was run in the Phenix GUI, you can select it from the drop-down menu labeled "Use partial solution from previous job" in the "Input and general options" tab. This will automatically load the solution. Note that unlike the CCP4i interface, Phaser-MR in Phenix does not make use of .sol files, since the information is passed internally. You should then load the new ensembles to search for; ensembles that were used in the previous run do not need to be re-loaded unless you wish to find more copies. If you have already provided complete information about the ASU contents correctly, this will not need to be changed.

How do I run a new search starting from an existing model in the correct orientation and translation?

This might apply if you have a refined model, or a solution from another program. In this case, you should add the model as an ensemble, and check the box "Ensemble is fixed partial solution". This will instruct Phaser to accept the input coordinates without modification, and skip to the next ensemble to search for. You do not need to specify a separate search for this ensemble, as it will be included automatically.

MRage

How do I tell the program to use a local PDB mirror instead of fetching entries remotely?

See the general FAQ list.

MR-Rosetta

Why does mr_rosetta bomb and say "error while loading shared libraries: libdevel.so: cannot open shared object file: No such file or directory"?

This may indicate that somewhere your system is defining the shared libraries that Rosetta needs, and these are for a place that is not where Rosetta expects them to be. You can try to ignore the previous definitions this

way:

If you are using the bash or sh shells:

```
export PHENIX_TRUST_OTHER_ENV=1
```

or csh (C-shell):

```
setenv PHENIX_TRUST_OTHER_ENV 1
```

in the script where you run mr_rosetta, or before you run it from the command line.

Why does mr_rosetta or mr_model_preparation bomb and say "RuntimeError: Cannot contact EBI DbFetch service"?

This could mean just what it says...but also it could mean that you are behind a firewall and there is a proxy server you need to go through. You can use the following command to specify the proxy server (replacing it with YOUR proxy server).

If you are using the bash or sh shells:

```
export HTTP_PROXY=proxyout.mydomain.edu:8080
```

or csh (C-shell):

```
setenv HTTP_PROXY proxyout.mydomain.edu:8080
```

Problems installing Rosetta? Here are some suggestions:

- Download rosetta_source and rosetta_database separately. The bundle doesn't contain all the files.

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- Compilation fails on RHEL 6.1 (gcc 4.5 issue) but sails on RHEL 5.5.

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- Make sure the environment is set correctly: \$PHENIX_ROSETTA_PATH must point to the Rosetta installation directory (where rosetta_source and rosetta_database sit). (Note there was a typo in documentation with ROSETTTA instead of ROSETTA; fixed now.)

Make sure the environment is set correctly: \$PHENIX_ROSETTA_PATH must point to the Rosetta installation directory (where rosetta_source and rosetta_database sit). (Note there was a typo in documentation with ROSETTTA instead of ROSETTA; fixed now.)

- all directories in the Rosetta installation must be made accessible to users: find \$PHENIX_ROSETTA_PATH -type d -exec chmod 755 '{}' \;

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```
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See the general FAQ list.

Reference

Phaser crystallographic software. A.J. McCoy, R.W. Grosse-Kunstleve, P.D. Adams, M.D. Winn, L.C. Storoni, and R.J. Read. J Appl Crystallogr 40, 658-674 (2007).

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Frequently asked questions for phenix.refine

refine.html

Note: questions specific to the GUI can be found in its documentation.

Contents

- General
- Optimization methods
- Targets and restraints
- Non-crystallographic symmetry (NCS)
- B-factors/ADPs/TLS
- Twinning
- Using R-free
- Interpreting results
- Hydrogens
- Miscellaneous
- Citations
- References

General

What is the difference between phenix.refine and phenix.real_space_refine?

phenix.refine is reciprocal-space refinement and is normally used for crystallographic data (X-ray or neutron).

phenix.real_space_refine is real space refinement and is normally used for cryo-EM data.

How can I make phenix.refine run faster?

There are currently two options for this:

- compile from source with OpenMP support, which parallelizes the Fast Fourier Transform (FFT) used in structure factor and gradient calculation
- specify the 'nproc' parameter (or equivalent control in the GUI), which parallelizes various grid searches by starting multiple Python jobs

The OpenMP parallelization is not particularly efficient, since the FFTs take less than half of the typical runtime; a speedup of 40% is usually the maximum. The process-level parallelization with 'nproc' is most useful when restraint weight optimization is enabled, since these procedures can be run as multiple separate processes. In these circumstances a speedup of 4-5x is possible. However, a run using default parameters will not significantly benefit from setting 'nproc'.

There are several limitations to these options:

- OpenMP support is incompatible with the Phenix GUI
- 'nproc' is incompatible with Windows
- these approaches are incompatible with each other

I set the number of processors to 8 - why is phenix.refine still only using a single processor?

As noted above, the standard parallelization primarily affects weight optimization. If this is not performed, most of the program will run serially.

What type of experimental data should I use for refinement?

Either amplitudes (F) or intensities (I) may be used in refinement (in any file format), but intensities will be used preferentially. Both anomalous and non-anomalous data are supported; there does not appear to be any particular benefit to model quality using one or the other with the default strategy. However, anomalous data may be used to refine anomalous scattering factors for heavy atoms, and anomalous difference map coefficients will automatically be created in the output MTZ file. For these reasons, anomalous data are recommended, but in general the R-factors will be similar.

Should I use the MTZ file output by phenix.refine as input for the next round of refinement?

The only time this is necessary is when you refined against a dataset that did not include R-free flags, and let phenix.refine generate a new test set. In this case, you should use the file ending in "_data.mtz" for all future rounds of refinement. You do not need to update the input file in each round, as the actual raw data (and R-free flags) are not modified.

I ran AutoSol to get a partial model that I now want to refine. Which data file should I give as input: the original .sca file from HKL2000, or the file overall_best_refine_data.mtz from AutoSol?

Always use the MTZ file output by AutoSol. This contains a new set of R-free flags that have been used to refine the model; starting over with the .sca file will result in a new set of flags being generated, which biases R-free.

Why does phenix.refine not use all data in refinement?

Reflections with abnormal values tend to reduce the performance of the refinement engine. These are identified based on several criteria (see Read 1999 for details) and filtered out at the beginning of each macro-cycle. You can prevent this by setting `xray_data.remove_outliers=False`.

How many macro-cycles should I run?

We recommend at least five to ensure convergence, but in some cases (especially poorly refined structures) considerably more may be required for optimal results. (The default is three macro-cycles due to speed considerations.)

How should I decide what resolution limit to use for my data?

This is a contentious subject, and not settled, although it is widely agreed that throwing away usable data simply to reduce the R-factors is not an acceptable practice. Traditional criteria for resolution limits include (among others) truncating the data at the resolution where the mean I/σ falls below 2, but this may exclude valuable data. See the documentation on unmerged data (especially the cited references) for more details. In general it may be useful to include additional, weaker data in refinement in the final stages, since the weighting performed by maximum-likelihood refinement prevents these from degrading the model, and may improve it in some cases. (Note that more data will also cause phenix.refine to take longer to run, which often imposes a practical limit on resolution.)

Is it okay to refine against data that have been modified by anisotropic scaling or ellipsoidal truncation?

There is no technical reason why this is impossible. However, modified data should be avoided for several reasons:

- phenix.refine already performs anisotropic scaling internally and will correct the output map coefficients for anisotropy, so the pre-scaling should be redundant.

- Because phenix.refine (like REFMAC) by default substitutes F-calc for missing reflections when calculating 2mFo-DFc and similar map coefficients, ellipsoidal truncation will result in eliminated high-resolution reflections being replaced by F-calc in the maps. This will always make the maps look better, because they are more biased to the model!
- Even if you do choose to refine against anisotropically scaled data, you should always deposit the original, unmodified data in the PDB upon publication.

Optimization methods

When should I use simulated annealing?

Simulated annealing (SA) is most useful early in refinement, when the model is far from convergence. Manually built models, or MR solutions involving significant local conformational changes, are common inputs where SA can improve over simple gradient-driven refinement. It is generally less helpful later in refinement, and/or at high resolution.

When should I use rigid-body refinement?

This typically only needs to be performed once after molecular replacement, unless you dock in additional domains later. Continuing to use rigid-body refinement in later runs will not improve your structure, and only adds to the runtime.

What happened to the old fix_rotamers option?

As of version 1.8.3, the real-space refinement strategy now incorporates both global minimization and local fitting; the latter is similar to fix_rotamers but much faster and incorporates backbone flexibility. Note that it will not run at very high or low resolution, or when explicit hydrogen atoms are present.

When is the ordered solvent method useful?

This works out to approximately 2.8Å as the low-resolution limit, depending on data quality. At atomic resolution (beyond approximately 1.2Å), it is useful early in refinement when no waters have been placed yet, but as the structure becomes more complete it may remove weaker, partial-occupancy waters, and it is unable to handle static disorder (alternate conformations).

Targets and restraints

When should I optimize the geometry and/or B-factor restraint weights?

This may be beneficial if the automatic weighting does not pick a good scale for the X-ray and restraint terms; this will often be recognizable by higher-than-expected bond and angle RMSDs. In general it rarely hurts to optimize the weights, and often results in a significantly better refinement, but it is several times slower than ordinary refinement unless you have a highly parallel system. However, we strongly recommend weight optimization in the final round of refinement, where it becomes essential to prevent overfitting.

How can I set the target weights manually?

Our usual response is: don't do this manually, use the automatic optimization instead. Although this takes significantly longer to run, in practice most users will spend an equivalent amount of time manually adjusting the weights by trial-and-error. If you are certain you need to have manual control, the parameters fix_wxc (for geometry restraints) and fix_wxu (for B-factor restraints) will set the weights.

Why doesn't phenix.refine use weight optimization even though I selected this option?

Because the weight optimization takes into account (and tries to limit) the difference between R-work and R-free, it will be automatically disabled if these statistics already very close (or if R-free is greater than R-work), as often happens at the start of refinement from an MR solution.

My resolution is X Angstroms; what should RMS(bonds) and RMS(angles) be?

This is somewhat controversial, but absolute upper limits for a well-refined protein structure at high resolution are typically 0.02 for RMS(bonds) and 2.0 for RMS(angles); usually they will be significantly lower. As resolution decreases the acceptable deviation from geometry restraints also decreases, so at 3.5 Angstrom, more appropriate values would be 0.01 and 1.0. We recommend using the POLYGON tool in the validation summary to judge your structure relative to others at similar resolutions.

Why does my output model have very poor geometry (RMS(bonds) and RMS(angles))?

This usually means that the automatic X-ray/geometry weighting did not work properly; this can sometimes happen if the starting model also has poor geometry. Optimizing the weight (optimize_xyz_weight=True, or equivalent GUI control in the "Refinement settings" tab) will usually fix this problem.

The RMS(bonds) and RMS(angles) in the output model are too low after weight optimization - how can I make them higher?

This is a common misconception about geometry deviations, based partly on anecdotal experience with other refinement programs. The target RMSDs come from looking at very accurate high-resolution small molecule structures, so they reflect the real variation that should occur in geometry. At lower resolution, even though you know that the bond lengths and bond angles must vary as much as they do in high-resolution structures, there isn't enough experimental data to tell in which direction they should deviate from the expected values. So it is reasonable for the refined RMSDs to be lower than the targets.

(Anecdotally, we have found that phenix.refine often refines to much tighter RMSDs with similar R-frees to other programs. This may reflect different approaches used in the geometry restraints, X-ray target, or optimization methods, but it should not be cause for concern.)

I have experimental phases for this structure, but the initial maps were poor. Should I still use the MLHL target?

The experimental phases used to restraint refinement describe a bimodal probability distribution for every angle, rather than the single values used to generate a map. In most cases the additional restraints will not hurt refinement, and can often help.

Why is phenix.refine messing up my ligand geometry?

This often happens when the restraints were generated using ReadySet from a PDB file, and the ligand code is not recognizable in the Chemical Components Database. eLBOW will try to guess the molecular topology based on the coordinates alone, but this is imprecise and may not yield the desired result. For best results, restraints for non-standard ligands should be generated in eLBOW using a SMILES string or similar source of topology information.

What can I do to make my low-resolution structure better?

In general, if NCS is present in your structure, you should always use NCS restraints at low resolution; it is worth trying both the Cartesian (global) and torsion restraints to see which works best for your model. This alone usually helps with the geometry and overfitting, although it is rarely sufficient by itself. There are also several different types of restraint specifically designed to help with low-resolution refinement (consult the full phenix.refine manual page for details on each):

- Reference model: this restraints the torsion angles to a high-resolution model, using a potential that gently releases angles which are genuinely different (allowing both local deformations or domain movements). This is usually the best option, and typically improves both R-free, overfitting, and validation statistics. However, it depends on the availability of a suitable reference model.
- Secondary structure restraints: these are simply harmonic distance restraints between hydrogen-bonding atoms in protein helices and sheets, and nucleic acid base pairs. Annotation can be either automatic or manual. The primary limitation is currently the need for outlier filtering, which removes spurious bonds but also some genuine ones. It has a small effect on validation statistics, and typically none on R-free, but it does keep secondary structures stable.

I have ions very close to water molecules/protein atoms, and phenix.refine keeps trying to move them apart. How can I prevent this?

Use `phenix.ready_set` or `phenix.metal_coordination` to generate custom bond (and optionally, bond angle) restraints, which will be output to a parameter file ending in ".edits". If you are using the PHENIX GUI, there is a toolbar button for ReadySet in the `phenix.refine` interface, which will automatically load the output files for use in `phenix.refine`.

I had previously generated custom restraints using ReadySet in the PHENIX GUI, but the atoms have changed. Now `phenix.refine` crashes because it can't find the atom selections. How do I remove the old custom restraints?

In the Utilities menu, select "Clear custom restraints."

What does "sigma" mean for geometry restraints, and what values are appropriate?

The sigma is the estimated standard deviation (e.s.d.) of the target value. For distance (bond) restraints, the sigma will be in Angstrom units; for bond angle and dihedral restraints it will be in degrees. A typical covalent bond will have a sigma of 0.02 or 0.03, and weaker "bond" restraints such as hydrogen bonds or metal coordination restraints will be looser with sigma between 0.05 and 0.1. For bond angle restraints, sigma is usually a few degrees; for dihedrals, up to 20-30 degrees is not uncommon. (See also the next question below.)

I want my ligand geometry to be absolutely perfect with no deviation from the target value(s). Can I just set the sigmas to zero or an extremely low value?

You cannot set the sigma to zero because the weight on the restraints is equal to $1/\sigma^2$. A very low value will not crash, but it will almost certainly confuse the minimizer and result in a sub-optimal structure, because those restraints will dominate the target and gradients, forcing the minimizer to take inappropriately large steps.

One part of my structure is particularly poor; how can I make the geometry restraints tighter for only those atoms?

The parameter scope `refinement.geometry_restraints.edits.scale_restraints` allows you to upweight the restraints for the specified atom selection, for any combination of bond lengths, bond angles, dihedral angles, and chirality. For instance:

```
refinement.geometry_restraints.edits.scale_restraints {
  atom_selection = "chain B"
  scale = 2.5
  apply_to = *bond *angle dihedral chirality
}
```

You may specify multiple such parameter blocks. The scale should be a relatively small number, typically less than 10 (you may also reduce the weight if you want). Note that since this affects the path of the minimizer,

the overall geometry RMSDs (and the deviations for restraints which were not scaled) will likely change as a result.

Can I make phenix.refine restrain the planarity of RNA/DNA base pairs?

At present this requires specifying each base pair individually as a custom planarity restraint (part of the parameter scope refinement.geometry_restraints.edits), which may be excessively time-consuming for large structures. Automatic planarity restraints may be added in a future version.

Can I use a reference model to restrain ligand coordinates?

The reference model restraints are only intended to work with macromolecules. However, you may use separate harmonic restraints for any subset of atoms; these will tether the selected atoms to their initial coordinates. This can be effective when you already have good geometry and map fit for the restrained atoms; however, it does not allow for genuine conformational differences.

How do I stop simulated annealing from pushing certain atoms too far out of density?

The harmonic restraints are suitable for this purpose. This is especially useful when generating a simulated annealing omit map, where atoms may move to fill voids left by omitted scatterers.

Non-crystallographic symmetry (NCS)

When should I use non-crystallographic symmetry (NCS) restraints?

This will of course require that NCS is actually present in your crystal. An approximate cutoff for NCS restraints is 2.0 Angstrom - at higher resolution the data alone are usually sufficient, but at lower resolution additional restraints are usually necessary.

What is the difference between global and torsion NCS, and which one should I pick?

The global NCS restraints groups as rigid bodies, where all atoms in each group are expected to be related to the others by a single rotation and translation operation. This does not respect local deformations in the related molecules, which are common even at lower resolution. The torsion NCS restraints restraint dihedral angles instead, and allow them to be unrestrained if genuinely different. This option has been made the default if NCS restraints are activated, since it usually results in significantly better refinement, and rarely performs worse than using no restraints at all.

How do I specify the .ncs_spec file from AutoSol or phenix.find_ncs for use in refinement?

The .ncs_spec files containing rotation and translation matrices are only used in density modification and model building. For refinement, the NCS relationships are always given as atom selections, and in the case of the default torsion NCS restraints, the automatically detected restraint groups should be very accurate.

How does one define NCS groups manually in the Phenix GUI?

If you are using the torsion NCS restraints, this should not be necessary, but you can enter atom selections manually by selecting "Detect NCS groups" from the Utilities menu. If you have identified a structure where manual group assignment is necessary for refinement to behave properly at low resolution, this may indicate a bug in phenix.refine; please contact us directly by emailing "help@phenix-online.org".

For the global/Cartesian NCS restraints, manual selections often are necessary, and the same menu item will open a different selection window if the global option is selected. However, we encourage you to try the torsion NCS restraints first before deciding to use the global parameterization.

Both NCS restraint types make my structure worse - what should I do?

If you are refining a structure at high resolution (better than 2.0 Angstrom), this is not unexpected; NCS restraints are usually unnecessary in such cases. At lower resolution, this may indicate a bug; we encourage you to contact us directly.

What impact do NCS restraints have on the electron density maps?

The maps will only be affected to the extent that the NCS-related chains will be relatively similar, and therefore the map phases will reflect this similarity. The maps will not be directly modified using the NCS relationships, however. If you want phenix.refine to perform NCS averaging on the maps, this is available as an option for individual map coefficients.

Do NCS restraints apply to B-factors?

By default, no; our testing has indicated that this is rarely beneficial and in a significant fraction of structures it may actually yield worse results (due to real differences in the disorder present in NCS-related copies). If you want B-factors to be restrained between NCS groups, check the box labeled "Restraining NCS-related B-factors" in the "NCS options" window, or on the command line, specify `ncs.restrain_b_factors=True`.

B-factors/ADPs/TLS

When should I use TLS?

TLS refinement is generally valid at any resolution; at low resolution, it may be best to make each chain a single group, instead of trying to split them into smaller pieces. However, it is best to wait until near the end of refinement to add TLS; until then you should refine with isotropic ADPs only.

Can I use both TLS and anisotropic ADPs?

Yes, but not for the same atoms - since TLS is essentially constrained anisotropic refinement, the two methods are mutually exclusive.

Where is the switch for anisotropic vs. isotropic B-factors/ADPs?

phenix.refine does not have a single global switch for defining ADP parameterization; rather, when the "Individual ADPs" strategy is defined, the program uses several criteria to determine how atoms should be treated:

- By default, atoms that are anisotropic in the input model (i.e. have ANISOU records) will be kept anisotropic if the resolution is at least 1.7Å; isotropic atoms remain isotropic.
- If the resolution is lower, all atoms will be converted to isotropic unless otherwise specified.
- The atom selections for isotropic and anisotropic atoms may also be defined explicitly. Typically, the hydrogens will always be isotropic; protein atoms will be anisotropic at high resolution, and sometimes waters as well.

In the GUI, several common parameterizations are pre-defined in the dialog for entering ADP selections. Note that although it is possible to combine all of the different ADP refinement strategies in a single run, the atom selections for individual and grouped refinement may not overlap, nor may the selections for anisotropic ADPs and TLS groups.

When should I refine anisotropic ADPs instead of TLS groups?

There is no precise cutoff where you should turn on anisotropic ADPs, but these are approximate guidelines, assuming that the data are actually complete to the indicated resolution:

- At 1.4 Angstrom resolution or better, refine protein/nucleic acid/ligand heavy atoms (C/N/O or heavier) anisotropically, and waters isotropically.

- At 1.1 Angstrom or better, waters should also be anisotropic.
- It is almost never appropriate to refine hydrogens anisotropically.

There may be circumstances where anisotropic refinement is permissible at slightly lower resolution, but 1.7 Angstrom is probably a lower limit. Exceptions may sometimes be made for metal ions, since they scatter very strongly. As always, you should use the drop in R-free to judge whether the change in parameterization was appropriate - a decrease of 0.5% (i.e. 0.005) or better indicates success.

When should I refine grouped B-factors/ADPs instead of individual?

It is again difficult to give an exact rule, since it depends on several properties of the crystal including resolution, solvent content, presence of NCS, etc. In general, the higher the data-to-parameter ratio, the more likely individual ADPs are to work well. As an approximate example, consider these two hypothetical structures:

- a 3.5 Angstrom structure of a single protein chain per asymmetric unit, with 38% solvent content
- a 4.0 Angstrom structure of 3 NCS-related chains, with 70% solvent content

In this case, the latter structure can probably be refined with individual ADPs, while the former is more marginal. If in doubt, early rounds of refinement may be done with grouped ADPs, switching to individual as the structure nears convergence. In general, it is usually worth trying individual ADPs at some point; ultimately the effect on R-factors (primarily R-free, but also the gap between R-work and R-free) is the most important guideline.

Twinning

When should I use twinned refinement?

You should only do this if Xtriage indicates that the intensity statistics are abnormal and cannot be explained by picking the wrong space group. The estimated twin-law-specific twin fractions displayed in Xtriage should not be used to determine whether twinning is present.

I performed twin refinement and my R-free went down by 1%; does that mean my structure is twinned?

No, because R-factors calculated with and without twinning are not necessarily on the same scale. In phenix.model_vs_data, the structure is only considered twinned if application of a twin law reduces R-work by at least 2%. Note that if you specify twin_law=Auto, phenix.refine will use the same procedure to determine the twin law (if any).

But what if the reported twin fraction is nearly 50%?

This often means that your data are under-merged, i.e. the crystal symmetry is too low and the applied twin operator is substituting for a symmetry operation. If this is case you should merge the data to higher symmetry and reduce the model to the new ASU. Of course it is also possible to have perfectly twinned crystals, but the high twin fraction output by phenix.refine does not by itself indicate twinning.

My data has multiple twin laws; can I use these in Phenix?

Currently we only support a single twin law; programs capable of refining tetartohedrally twinned structures are REFMAC and SHELXL.

What are the disadvantages of twinned refinement?

In Phenix specifically, twinned refinement uses a least-squares (LS) target instead of the more powerful maximum likelihood target used for conventional refinement. Additionally, twinned refinement makes no use of experimental phases (if available) as restraints. Some refinement protocols may not work with twinning,

although as of April 2013 most of these have been fixed. More generally, the output map coefficients will have a significantly worse model bias problem than conventional maps; this effect increases as the twin fraction nears 0.5.

Using R-free

phenix.refine stops with this error: "R-free flags not compatible with F-obs array: missing flag for 100 F-obs selected for refinement". How do I fix this?

This means that the experimental data array contains reflections that are not present in the R-free flags array (this is required even if the flag is False). Use the reflection file editor to extend the existing R-free flags to cover all reflections in the file. Make sure that the checkbox labeled "Extend existing R-free array(s) to full resolution range" is checked.

phenix.refine stops with an error message about the model being refined against a different set of R-free flags. How can I fix this?

First, make sure that you have not actually generated a new set of R-free flags; once you have these flags for a given dataset, you should continue using them throughout the building and refinement process. The error message is intended to guard against this happening accidentally. If, however, you have collected new higher-resolution data and extended the old R-free flags, then the error message may be ignored. The R-free flag comparison is based on information stored in the REMARK records in the input PDB file, so if you edit the PDB file and remove the line containing the word "hexdigest", the refinement will be able to continue.

I have a model that was previously refined against a previous set of R-free flags that I don't have access to. How can I avoid biasing the R-free when I refine this model?

There are several methods for this, but the easiest is to reset the B-factors (using PDBTools or the "Modify start model" option in phenix.refine) and run simulated annealing on the coordinates. If you are especially worried about bias you can alternately randomize the coordinates and perform energy minimization, or build an entirely new model starting from the phases calculated from the original model. However, we usually find that the annealing is aggressive enough to remove any "memory" of the original R-free flags.

My resolution is X Angstroms, and my R/R-free are Y and Z. Am I done refining?

A partial answer can be obtained by looking at POLYGON, which plots histograms of statistics for PDB structures solved at similar resolutions, and compares these to the statistics for your output model. As a general rule, R-factors alone should not be used to decide if a structure is "done", but should be examined in combination with the validation report.

My resolution is X Angstroms, the structure is complete and well-validated, the maps look great, but my R and R-free are still really high. How can I make them lower?

There are several possible explanations for this:

- Twinning with a relatively small twin fraction (perhaps 10%) may not obviously affect map quality, but can still have a significant impact on R-factors. Run Xtriage to look for evidence of twinning, and possible twin laws for using in phenix.refine. (As noted above, however, only one twin law may be used at a time.)
- Undiagnosed translational NCS (AKA translational pseudosymmetry) can have a similar effect. In this case, the diffraction images may have been processed incorrectly, thus missing the fainter spots that result from tNCS. The program LABELIT, or more specifically the command labelit.index (distributed with PHENIX, but not yet available in the GUI) can be run on the original images to provide appropriate indexing parameters.

The gap between R-work and R-free is very large - how can I fix this?

Overfitting during refinement is usually helped by adding more restraints, and/or tightening the standard geometry restraints. If the output geometry is already within reasonable limits (typically RMS(bonds) < 0.016 and RMS(angles) < 1.8), ideas to try include adding NCS restraints if NCS is present, secondary structure restraints, or reference model restraints (if a high-resolution structure is available). At lower resolutions (worse than 3.0Å), it may also be prudent to try grouped ADP refinement, and if desperate, Ramachandran restraints. TLS refinement can often improve overfitting across a wide range of resolutions. However, depending on the degree of overfitting, it may be necessary to perform extensive manual rebuilding first. (Note that if the large R/R-free gap suddenly appears after refinement of a model that was previously not overfit, this usually indicates incorrect parameterization of the refinement, e.g. using anisotropic ADPs at an inappropriate resolution.)

I ran a round of refinement, rebuilt in Coot, and refined again. My previous R-free was 0.25, but the new refinement starts out at 0.35. Why is it so high?

The initial R-factors reported by phenix.refine are without bulk-solvent correction, which usually has a significant impact on R-factors. Once this step is performed (at the start of the first macrocycle), the R-free should drop immediately to approximately the expected value.

Why does phenix.refine give me a different R-factor than program X?

There are many explanations for this. Even without minimization, the bulk solvent and scaling methods alone may account for as much as a 1% difference or more in calculated R-factors. For refinement results, the differences in target functions, restraints, and minimizers may be significant. In some cases the explanation is as simple as running too few cycles of refinement for one or the other program. In general, if you find a case where phenix.refine performs significantly worse than another program, we encourage you to contact us at help@phenix-online.org.

Are R-free flagged reflections included in the maps used for real-space refinement?

No, this is almost guaranteed to bias R-free; these reflections are removed internally prior to map calculation. However, the output maps will include these reflections unless you explicitly request otherwise.

Interpreting results

I solved my structure by MR and refined, and the R-free is 45%. The maps are messy and I see a lot of difference density, but none around my molecule. Why isn't refinement working?

This frequently indicates that too few copies of the structure were placed by MR, and an additional chain needs to be added. Remember that the predictions of unit cell contents based on the Matthews coefficient (performed by Xtriage, for example) only provide an estimate, not an exact answer. At high resolution a solvent content of 40% or less is quite common.

Why am I seeing negative blobs in the difference (mFo-DFc) map in hydrophobic voids?

In previous versions of Phenix, the bulk solvent mask was often being extended to include these regions. This should no longer be a problem as of July 2013, but if you continue to see this effect, please contact us at help@phenix-online.org.

After coordinate and B-factor refinement the heavy atoms in my structure have negative mFo-DFc peaks. How do I get rid of these?

Refining the occupancies will often fix this problem. Alternately, if a significant amount of anomalous scattering is expected at the wavelength used for data collection, anomalous group refinement may also be

helpful. We do not recommend setting the B-factor manually and turning off refinement for the problematic atoms.

After refinement the mFo-DFc map has positive density around correctly placed atoms that are already at full occupancy. Is my model missing something?

Usually this means that the initial B-factors of the input model were too high for refinement to converge. Typically the minimizer is very good at raising low B-factors to the correct value, but gets stuck in the opposite direction. The observed result could happen if you refine starting from a lower-resolution model, or if you build new residues in Coot and the default B-factors are well above what they should be (typically this only happens at atomic resolution). To fix the problem, you just need to reset the B-factors to a suitably low value. This can be done at the start of refinement by setting the parameter `refinement.modify_start_model.adp.set_b_iso`; in the GUI, this can be found in the Settings menu under "Modify start model" --> "Modify ADPs...".

Why does Phenix validation report a different number of Ramachandran outliers than Coot/Procheck/the PDB/other program?

The phi/psi distributions used in Phenix are the same as those in the MolProbity server (Chen et al. 2010), and are based on a curated set of 8000 high-resolution crystal structures. There are now six distributions for different residue classes (general, glycine, Ile/Val, pre-Pro, cis-Pro, and trans-Pro). These distributions are stored in 2-degree increments. Other programs generally use older and/or less precise distributions to score phi/psi angles, which frequently results in disagreements for residues which are on the border of allowed and outlier regions of the plot. We suggest that you rely primarily on the results in Phenix (or MolProbity), as the distributions we use are very accurate and based on the latest structural data.

How come I have a bunch of clashes with water molecules in the validation results after running solvent update?

phenix.refine is relatively aggressive in placing solvent atoms in unmodeled density. However, this may sometimes result in clashes if the density represents ions, unmodeled residues, or alternate conformations rather than solvent. For this reason, we recommend that the final round of refinement not include solvent update (regardless of resolution), after any clashing water atoms have been removed. (You should also attempt to model the observed density features if possible, although this is not always straightforward.)

What does it mean when a water atom lies in a positive mFo-DFc peak?

This indicates that the water is actually something heavier. If the density shape (for both 2mFo-DFc and mFo-DFc maps) is still relatively isolated and spherical, this suggests an ion of some sort (chloride, calcium, etc.). You can use the built-in ion identification in phenix.refine to try to model it, or make the assignment manually based on examination of the local environment. We also recommend looking at the anomalous difference map, as many ions will have significant anomalous scattering at typical data collection wavelengths. Alternately, if the density is more extended, this indicates that a larger ligand is bound.

I still have some mysterious mFo-DFc blobs that I can't identify; what should I do with them?

One option is to use the ligand identification tool to try to place various common small molecules in the density; we recommend being very conservative in filtering the results. It is very helpful to check the conditions used for purification, crystallization, and cryoprotection, since these frequently result in buffer components being visible in the maps. However, if you are unable to positively identify blobs, we recommend leaving them empty (and perhaps also noting this in the deposition remarks or publication). We do not recommend filling them with water molecules, as this misrepresents the true identity of the blobs, nor do we recommend using "unknown" atoms as placeholders (especially since these are incompatible with most tools in Phenix).

Hydrogens

When should I refine with hydrogens?

This is largely a matter of personal preference. Using explicit riding hydrogen atoms can improve geometry at any resolution; at higher resolutions, approximately 2 Angstrom or better, they will generally improve R-free as well. At atomic resolution (1.5 Å or better) they should always be part of the final model. Note that at unless you have true subatomic resolution (0.9 Å or better), the hydrogens should always be refined as "riding", meaning that their coordinates are defined by the heavy atoms, not individually refined against X-ray data.

How can I tell phenix.refine to add hydrogens to my model?

The command-line program does not add hydrogens; this is performed by a separate program, phenix.ready_set. However, in the GUI there is an option in the "Refinement settings" tab to add hydrogens, which simply runs ReadySet internally immediately before starting phenix.refine.

What about water molecules?

Although phenix.ready_set includes an option to add hydrogens to waters, we do not recommend this unless you have exceptionally high resolution and/or neutron data.

Why are my hydrogen atoms added by PHENIX exploding when I run real-space refinement in Coot?

Versions of Coot prior to 0.6.2 used a version of the CCP4 monomer library with hydrogen atoms named according to the PDB format version 2 standard; PHENIX can recognize these, but defaults to PDB v.3. To reconcile the different conventions, you can download the newer version of the monomer library (currently available [here](#)) and set the environment variable COOT_REFMAC_LIB_DIR to point to the directory in which you unpack it. However, newer versions of Coot do not appear to have this problem.

I refined with riding hydrogens - why are they present in the output model?

phenix.refine will always output all atoms that are part of the atomic model, regardless of how they were parameterized during refinement. Use of the "riding" model does not necessarily guarantee that the hydrogen positions and B-factors are reproducible, as they differ between various programs (or program versions).

Why can't PHENIX automatically remove hydrogens from the output PDB file?

We strongly discourage removing any atoms used in refinement from the model, as it makes reproducing the published R-factors very difficult and eliminates essential information about how the structure was refined.

How are X-H (hydrogen to heavy atom) bond lengths defined, and why are these different than the ones in molecular mechanics programs?

Since X-rays are diffracted by electrons, the default hydrogen positions are defined as the centers of the electron clouds, not the atomic nuclei. In most cases this results in the bond lengths being 0.1 Å shorter than they would be if the distances to the nuclei were given. At high resolution, this will more accurately model the X-ray data. The VDW radii used for nonbonded restraints will be larger to compensate for the shorter bonds.

If you are performing neutron refinement, the nuclear positions will be used instead, resulting in longer bond lengths and smaller VDW radii. The MolProbity validation should handle both scenarios appropriately.

Miscellaneous

How can I model a charged atom?

The charge occupies columns 79-80 at the end of each ATOM or HETATM record, immediately following the element symbol. The format is the number of electrons followed by the charge sign, for example "1-" or "2+".

You can edit the PDB file manually to add this, but we recommend using phenix.pdbtools:

```
phenix.pdbtools model.pdb charge_selection="element Mn" charge=2
```

This is also available in the GUI under "Model tools". The effect of setting the charge will be to use modified scattering factors for X-ray refinement, which can be helpful if you notice difference density appearing at ion sites. Note that it will have no effect on the geometry, since phenix.refine does not take electrostatics into account.

I can't see density for an arginine sidechain beyond the C-gamma atom. How should I model it?

Opinion in the crystallography community differs on the proper approach to disordered sidechains, with significant support for both of the following methods voiced on the PHENIX and CCP4 mailing lists:

Delete all atoms not visible in density, but leave the residue name alone. This is arguably the most conservative approach, as it avoids modeling any features not supported by the data, and it is consistent with the treatment of missing loops. The main disadvantage is aesthetic, since it is more difficult to visualize and interpret the biological effects of a structure with missing sidechains. Anecdotal evidence suggests that some non-crystallographers may be confused by this. Pick an appropriate rotamer, and let the B-factors rise to account for disorder. This avoids truncated sidechains that may be mistaken for other residues, and is more realistic when interpreting surface electrostatics. The atomic B-factors and coordinates are actually refined against the data, however weak. It is potentially dangerous because it implies a greater level of confidence in these positions than is justified by the data. Additionally, the ADP restraints will keep the B-factors of nearby atoms similar (within some tolerance), which is normally essential for stable refinement but may artificially lower the B-factors of disordered sidechains.

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A third approach, setting the occupancy of missing atoms to zero but leaving them in the model, is strongly disfavored, as the resulting positions and B-factors are entirely theoretical (but not immediately obvious as such).

Running phenix.model_vs_data (or the validation GUI) results in a slightly different R-factor than reported in the PDB header by phenix.refine. Shouldn't these be the same?

phenix.refine and phenix.model_vs_data use the same code to perform the bulk solvent correction and scaling, so they should report approximately the same R-factors given identical inputs. The discrepancy arises when taking a PDB file from refinement and running it back through phenix.model_vs_data. Because of the limited precision of the PDB format (three digits after the decimal point for coordinates, two digits for B-factors and occupancy), the atomic properties recorded in the PDB file will not be exactly the same as their actual refined values. In practice the difference in R-factors is statistically insignificant, however.

When should I perform anomalous group refinement?

This is most useful when you have a large number of strong anomalous scatterers, where map artifacts are common. In such cases a more precise modeling of atomic scattering may improve the R-factors as well. It is generally not necessary for routine cases, but it may be advantageous for identifying weak anomalous scatterers (such as ions from the crystallization condition) when used to calculate an anomalous log-likelihood gradient (LLG) or residual map at the end of refinement.

How does phenix.refine deal with atoms on special positions?

There are two ways to handle atoms on special positions (e.g. symmetry axes):

If the occupancy is set to the value expected for the special position (e.g. 0.5 for an atom on a two-fold axis) or less, the coordinate position will be refined, and it will not interact with its symmetry mates. If the occupancy is set to 1, the atom will be constrained to stay at the special position, and the occupancy will be corrected internally when calculating structure factors.

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- If the occupancy is set to 1, the atom will be constrained to stay at the special position, and the occupancy will be corrected internally when calculating structure factors.

Note that the partial-occupancy atoms on special positions will have their occupancies defined if using default settings; you can disable this by instructing phenix.refine to remove a specific atom selection from occupancy refinement (the keyword `refinement.refine.occupancies.remove_selection`, or in the GUI, edit the atom selections for the Occupancy strategy).

I have sidechains in multiple conformations interacting across a symmetry axis, where the 'A' conformations clash with each other and are moved out of density by refinement. How can I tell phenix.refine to ignore this interaction?

This is not uncommon (a classic example is Tyr378 in PDB ID 1GWE), but it is not automatically handled right now and the PDB format does not provide a means to label the atoms in such a way that clashes are avoided. The solution in phenix.refine is a parameter defining an atom selection for which nonbonded restraints should be disabled:

```
custom_nonbonded_symmetry_exclusions = "chain A and resseq 378"
```

How can I extract the isotropic B-factor equivalent from a structure refined with TLS or anisotropic atoms?

You don't need any extra steps; the B-factor column in ATOM records in the PDB (or mmCIF) file will already be the total B-factor.

Why does phenix.refine output ANISOU records for individual atoms even though I only performed isotropic and TLS refinement?

TLS refinement is essentially constrained anisotropic refinement, so the individual atoms are anisotropic (just not independent); the ANISOU records simply make this explicit, since they have a standard format and are recognized by a variety of programs, unlike TLS information in the PDB header.

Why doesn't the PDB header report the bulk solvent parameters `K_sol` and `B_sol`?

Newer versions of Phenix use an improved bulk-solvent correction and scaling procedure which uses an entirely different parameterization that we find performs better (Afonine et al. 2013; see also Uson et al. (1999) Acta Cryst. D55, 1158–1167).

What is the difference between the various scattering tables? When should I use something other than the default?

If you are refining a neutron structure, you should of course use the neutron scattering table. The other tables are all X-ray-specific; the default, `n_gaussian`, is the best to use, as it uses dynamically defined number of Gaussians to approximate tabulated form-factors with required accuracy. `it1992` is commonly used in other programs - this is four Gaussians plus constant, taken from International Tables 1992 edition. `wk1995` is from (Waasmaier & Kirfel 1995), which is five Gaussians and is more accurate (but slower) than `it1992`.

Why doesn't `phenix.refine` use a constant number of resolution bins/why doesn't `phenix.refine` divide reflections evenly into resolution bins?

Binning reflections evenly is inappropriate for the bulk solvent correction and scaling method used in `phenix.refine`, which instead uses logarithmic binning. The logic is explained in Afonine et al. (2013):

"This scheme allows the higher resolution bins to contain more reflections than the lower resolution bins and more detailed binning at low resolution without increasing the total number of bins. An additional reason for using logarithmic binning is that the dependence of the scales on resolution is approximately exponential (see previous sections), which makes the variation of scale factors more uniform between bins when a logarithmic binning algorithm is used."

Citations

How should I cite `phenix.refine`?

Either (Afonine et al. 2012) or (Adams et al. 2010) is suitable; we recommend the later if you used additional components of Phenix. If you used the integrated MolProbity validation in the GUI, you should also cite (Chen et al. 2010) and/or (Echols et al. 2012).

How should I specify the refinement program in my PDB deposition?

If the PDB or mmCIF file you are depositing was output by `phenix.refine`, the refinement program (including version number) is already specified in the header. Otherwise, "PHENIX (version 1.8.4)" is suitable (replaced with the actual version number). However, note that if you used multiple refinement programs (for example REFMAC and PHENIX) during the course of structure determination, only the last will usually be named in the header, so we suggest that you edit this information during deposition to complete the list.

What are some references for the underlying methods used in `phenix.refine`?

For technical background, the most thorough source is (Afonine et al. 2012), which contains all of the relevant citations. We recommend that everyone who uses `phenix.refine` read this paper at some point even if you are not concerned with theory, as it provides a more detailed explanation for the methods and motivations behind the program than this documentation.

Where can I read more about the principles of refinement in general?

Bernhard Rupp's "BioMolecular Crystallography" is the most modern and complete reference, and includes a detailed explanation of the maximum likelihood methods used in `phenix.refine` and many other programs.

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